DISTILLATION SCIENCE

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This text was compiled on 07/13/2026

TABLE OF CONTENTS

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1: Overview

This is **Part I, Overview** of a ten-part series of technical articles on Distillation Science, as is currently practiced on an industrial level. It is organized as it would be used for the industrial process design of distillation columns, which may differ from the way it is introduced to students academically. As a part of distillation column design, these articles also involve the vapor-liquid equilibria (VLE) of binary systems, with potential extension to multi-component systems.

It is assumed that the reader has a basic understanding of distillation, as a means of separating two or more volatile fluids by the difference between their boiling points; and also the purpose of a distillation column's various components. A variety of texts exist that explain these concepts on a basic level. From such pre-requisites, the purpose of this series of articles is to expand these basics to the degree necessary for commercial applications with real fluids - which typically do not follow ideal behavior. It is also assumed that the reader is familiar with the concepts of molar units, mole fractions, the bonding of chemical elements, vapor pressure, latent heat of vaporization, and a compound's critical point - all of which are found in college freshman-level textbooks. In discussing mathematical relationships, it is assumed that the concepts of differentiation and integration are known to the reader.

While chlorosilanes and their electronic impurities are used as recurring examples, the technology developed here has great application in other mildly polar and hydrogenated compounds, which are normally excluded from theoretical treatment in many texts. The chlorosilane homologue starts with silane (SiH_4), and concludes with silicon tetrachloride ($SiCl_4$), as the Si-H bonds are incrementally swapped for Si-Cl bonds.

Chlorosilanes (and many of the electronic impurities) are non-naturally occurring compounds, but are key to the manufacture of high-purity silicon (solar photovoltaics and modern electronic integrated circuits, aka “computer chips”); as well as silicon-based chemicals such as silicones and organic/inorganic coupling agents. Applications for distillation science are found in both bulk separations of these fluids, as well as the purification by high-reflux processing, to parts-per-billion levels.

Furthermore, this technology is applicable to a wide variety of other non-organic fluids such as refrigerants, biologic pre-cursors, and pharmaceutical pre-cursors (i.e., not the final biological or pharmaceutical product, but rather the compounds used as catalysts and building-block agents for assembling these specialized chemical products).

To make the discussion of Distillation Science more generic, temperature, pressure, and molar volume (T, P, and V) are often expressed in dimensionless reduced units, designated respectively as T_r , P_r , and V_r. $T_r = T/T_c$; $P_r=P/P_c$; $V_r = V/V_c$. By definition, T_r =1, P_r =1, and V_r =1 simultaneously at the critical point. When temperature and pressure are given as T and P, these are in absolute units of Kelvins and atmospheres. For those more familiar with pressures in other absolute units such as PSIA, bar absolute, and Pascals absolute, converting those units to absolute atmospheres is easily done using internet-accessible conversion tables and calculators.

- **Part II, Existing Vapor Pressure Equations** deals with the pure component VP relationships commonly found in textbooks, as well as their limitations. This article lays the basis for subsequent articles.
- **Part III, Critical Properties and Acentric Factor** deals with the tabulation of these properties for selected fluids based on globally collected data. Data analysis and validation are discussed, as well as estimation techniques for those fluids for which data is either poor or non-existent. Critical properties are used to convert temperature, pressure, and specific volume from conventional units used in both chemistry and chemical engineering, to the reduced form of **Part IV**. In **Parts V** and **VII**, the value of the acentric factor is important.
- **Part IV, New Vapor Pressure Equation** expands on the basics of **Part II** and the results of **Part III**, to show how a new vapor pressure equation allows the practice of distillation applications at the elevated pressures more common to industry. The culmination of this article is a thermodynamically consistent equation that is valid between the atmospheric boiling point and the critical point, and which allows evaluation of other required distillation properties such as saturated phase densities and latent heat of vaporization.
- **Part V, Equation of State** deals with the recommendations for best EOS, as well as the mathematical techniques for solving such non-intrinsic EOS equation forms.
- **Part VI, Fugacity** deals with the departure of apparent pure-component vapor pressure in binary mixtures, as is commonly found in practical application of distillation science. The equations for evaluating fugacity coefficents are given.
- **Part VII, Liquid Activity Coefficients** deals with the application and estimation of Liquid Activity Coefficients, as commonly found in practical application of distillation science. Various Liquid Activity Coefficient models are reviewed along with some limited data and a recommended estimation technique given where data is lacking.

- **Part VIII, VLE Analysis Methods** discusses the recommended methodology used when data-collecting binary systems, to assure that systemic errors are minimized. This topic also deals with validation whenever data collection is done on reactive fluids, which can disproportionate or dimerize during study.
- **Part IX, Putting It All Together** shows how to combine the components of the above distillation science articles in a practical application.
- **Part X, Convergence Strategy** illustrates how solutions are best obtained for additional fluids using non-intrinsic or nested-loop equation methods, such as the recommended vapor pressure equations. While this topic is more mathematical or computer-science oriented than chemistry-centered, it is a necessary technique to understand when dealing with modern technology. No complex mathematics are required.

Equation Notation Used

Notation	Usage	Units	Notation	Usage	Units
NBP	Normal Boiling Point	°K	q	Watson exponent	-
P, VP	Pressure, Vapor Pressure	atmospheres	ΔH_{vap}	Latent heat of Vaporization	Cal/g-mol
P_c	Critical Pressure	atmospheres	ΔH_{vb}	Latent heat @ T_b	Cal/g-mol
P_r	Reduced Pressure	-	T-S ΔH_{vb}	Thek-Stiel VP equation parameter	Cal/g-mol
R	Gas constant	Cal/g-mol°K	Z	compressibility	-
T	Temperature	°K	Z_v & Z_L	compressibility of saturated vapor & liquid	-
T_b	Temperature @ Normal Boiling Point	°K	ΔZ_{vap}	(Z_v-Z_L) @ vaporization	-
T_c	Critical Temp	°K	α	Riedel derivative	-
T_r	Reduced Temp	-	α_c	α @ critical	-
T_{br}	Reduced T_b	-	ψ	Vapor Pressure derivative	-
V_c	Critical Volume	cc/g-mol	ø	fugacity coefficient	-
k	H bonding parameter	-	γ	liquid activity coefficient	-
k_{ij}	Binary Interaction Coefficient	-	ω	acentric factor	-

Units

The units used in these articles are temperatures in °K, pressures in atmospheres (absolute), and molar volumes in cc/gram-mole. Molecular weight (MW), and compressibility (Z) are dimensionless. For the above units, the value of the gas constant (R) is 82.057 atm-cc/mole-°K.

2: Vapor Pressure

Distillation Science (a blend of Chemistry and Chemical Engineering)

This is **Part II, Vapor Pressure** of a ten-part series of technical articles on Distillation Science, as is currently practiced on an industrial level. See also **Part I, Overview** for introductory comments, the scope of the article series, and nomenclature.

Part II, Vapor Pressure deals with the existing pure component vapor pressure (VP) equations normally found in textbooks. The content of this article is referred to in subsequent articles. The goal of this article is to explain the limitations of these less sophisticated VP equations, to show how best to use them, and to set up the introduction of a superior VP equation form in **Part IV**.

The original Clausius-Clapeyron equation relating VP, temperature, vaporization molar volume change (ΔV_v) and latent heat of vaporization (ΔH_v) dates back to mid-19th century and is derived from thermodynamic principles. The derivation is given in many college freshman-level texts and results from thermodynamic equilibrium between liquid and vapor phases. First, the differential of pressure with respect to system temperature (for both vapor and liquid) is re-arranged to:

$$\frac{dP}{dT} = \frac{\Delta H_v}{T \Delta V_{vap}} \tag{2.1}$$

If **(assumption 1)** the vaporization molar volume change (ΔV_{vap}) is set equal to the saturated vapor volume (V) by assuming the boiling liquid's molar volume is essentially zero; and if **(assumption 2)** the "Ideal Gas Law" (PV=RT) can hold for this saturated vapor, then Equation 2.1 in differential form the relationship becomes:

$$\frac{d(\ln P)}{d(1/T)} = \frac{\Delta H_{vap}}{RT} \tag{2.2}$$

When integrated, this results in the simplified Clausius-Clapeyron relationship that is found in most textbooks. However, there are limited conditions for which the above two assumptions are nearly correct: non-complex molecular structure and very low pressure (say, below atmospheric or very near atmospheric). For most compounds, and for most pressures encountered in normal industrial processes, neither of these assumptions holds very well and accuracy gets increasingly worse as pressure increases beyond atmospheric.

For real compounds and pressures normally encountered in industry, a term is added to the "Ideal Gas Law" ($PV = RT$) called **compressibility**, Z; so then the relationship becomes:

$$PV = ZRT \tag{2.3}$$

where compressibility Z is a function of that fluid's pressure, temperature, and other physical properties such as discussed in the **Part III**. Note that Equation (2.3) holds for both vapors and liquids: with Z_v being vapor compressibility, Z_L being liquid compressibility, and ΔZ_{vap} being the change in compressibility with vaporization. Now Equation (2.1) can be transformed into a more usable relationship for all fluids and all sub-critical pressures. In differential form, it is:

$$\frac{d(\ln P)}{d(1/T)} = \frac{\Delta H_{vap}}{\Delta Z_{vap} RT} \tag{2.4}$$

After integration between close temperatures T_1 and T_2 (so the ratio $\Delta H_{vap} / \Delta Z_{vap}$ can be taken as constant over that close range), Equation (2.4) becomes:

$$\ln(P_2/P_1) = \frac{\Delta H_{vap}}{\Delta Z_{vap} R} \times (1/T_1 - 1/T_2) \tag{2.5}$$

In order for this equation to be fairly accurate, it important for T_1 and T_2 to be close, since both ΔH_{vap} and ΔZ_{vap} are actually functions of temperature. Also note that if ΔZ_{vap} is set to unity, Equation (2.5) becomes the same as the simplified integrated Clausius-Clapeyron equation:

$$\ln(P_2/P_1) = \frac{\Delta H_{vap}}{R} \times (1/T_1 - 1/T_2) \tag{2.6}$$

It can now be understood that ΔZ_{vap} is the measure of the net effect of removing above assumptions **(1)** and **(2)** that allowed Equation (2.2). For most fluids near atmospheric pressure, ΔZ_{vap} is about 0.95 - 0.99, but as pressures increase toward a fluid's critical point (where vapor and liquid merge at a singularity), ΔZ_{vap} goes to zero. (But of course, ΔH_{vap} also goes to zero at the critical point, so the ratio $\Delta H_{vap} / \Delta Z_{vap}$ becomes undefined at this singularity).

The evaluation of ΔZ_{vap} normally requires use of an Equation of State (EOS), which is discussed in **Part V**. Using such an EOS to assess ΔZ_{vap} would allow vapor pressure vs temperature data to exactly align with ΔH_{vap} values. In **Part IV**, Equation (2.4) will be further developed and integrated without use of an EOS, up to pressures close to critical (usually about 95% of critical pressure in pure component systems).

Returning back to the above integrated form of Clausius-Clapeyron, Equation (2.5) suggests the commonly used empirical means to curve-fit VP data over small temperature/pressure ranges:

$$Ln(P) = A - B/T \tag{2.7}$$

The term "B" does not have any exact scientific significance, but works as a curve-fitting parameter and is normally shown with a negative sign, so as to have a positive value. Since Equation (2.7) has a limited range of application to low pressures, it can be improved for use near ambient and slightly higher pressures with an empirical form for curve-fitting VP vs T, called the *Antoine Equation*. However the constants A, B, and C have no scientific basis either and the equation form can still only be used over modest ranges (and low pressures).

$$\ln(P) = A - \frac{B}{T+C} \tag{2.8}$$

Additional constants D, E, F can be added to make the “extended Antoine” relationship for empirical curve-fitting; however there is still no correlation between the constants and scientific meaning.

$$\ln(P) = A - \frac{B}{T+C} + D \times T + E \times T^2 + F \times Ln(T) \tag{2.9}$$

From an industrial distillation column design perspective, even the extended Antoine VP relationships found in handbooks are not entirely adequate: they rarely reproduce the correct VP values at T_b, T_c and at T_r=0.7 (where the acentric factor is determined). That inadequacy undermines attempts to use modern Equations of State (EOS are discussed in **Part V**), forcing the use of the antiquated Van der Waals EOS. With any EOS, the extended Antoine VP relationship cannot connect latent heats and saturated (vapor and liquid) phase densities to vapor pressure. Additionally, none of these empirical VP equations reproduces the inflection point required by thermodynamics of a real fluid's VP vs T plot, normally occurring between T_r= 0.7 and Tr = 0.85. For economic reasons, many industrial processes operate at these higher pressures, so using empirical VP equations can lead to poor distillation column design. This is especially true in modern industrial applications where complex computer programs have automated several aspects of distillation column design.

In designing such an industrial-level distillation column, the various process simulation packages (e.g., ASPEN, VMG, etc) would then be fed with rather poor VP estimations. It is not uncommon with such simulation software to become non-convergent or to have the problem solution get stuck on a singularity. In the vernacular of the early days of computing, “Garbage in – Garbage out”.

The solution to this quandary is to find a better VP equation form that possesses all the criteria lacking in the VP equations shown in this article. A solution to that is proposed in **Part IV**.

However, Equation (2.7) and Equation (2.8) are not without merit, as long as they are used for the purpose they were derived: only for narrow ranges of temperature and at moderate pressures. Equation (2.8) (Antoine) is especially useful in correlating data taken around the atmospheric boiling point, T_b, to get a more accurate value from several experimental data points, rather than just one. It also allows data from several sources to be inter-compared, as long as they had a similar temperature range.

While evaluating the “A, B, and C” of Equation (2.8) may seem daunting, the solution is readily managed using some algebraic manipulation and a spreadsheet multiple regression function (e.g., MS Excel). Equation (2.8) is re-written as the algebraically equivalent

$$\ln(P) = A + \frac{AC - B}{T} - C \times Ln(P)/T \tag{2.10}$$

Then VP vs T data (always as absolute pressure and temperature) are converted into three columns for each data set: the dependent variable is “Ln(P)”, and the two independent variables are “1/T” and “Ln(P)/T”. When the regression is run, the regression's intercept will equal Equation (2.8)’s “A”; and the second independent variable’s coefficient (i.e., for Ln(P)/T) will equal the negative of Equation (2.8)'s "C". The value of Equation (2.8)'s "B" is determined algebraically from the first independent variable's coefficient (i.e., for 1/T)= (AC-B). See an example of using this solution procedure in below Table 2-1.

This curve-fit parameter solution technique does not work with Equation (2.9), so often the "Extended Antoine" equation is stated differently as

$$Ln(P) = A - \frac{B}{T} + C \times Ln(T) + DT + ET^2 \tag{2.11}$$

requiring a multiple linear regression (such as with MS Excel) be done with the four "independent" variables: 1/T, Ln(T), T, and T^2. Assuming the temperature range is narrow, the curve-fit quality is almost always statistically better with Equation (2.8) than with Equation (2.11), since there are two fewer constants. If the range is broader, such as spanning from atmospheric to the half-way point of critical pressure, then Equation (2.11) will give the better curve-fit.

When using experimental data to determine the normal boiling point (T_b), Equation (2.8) seems to work best. When using experimental data to determine the VP at T_r= 0.7 (i.e., evaluating the acentric factor), or inter-comparing VP measurements over a similar broader range, Equation (2.11) is preferred. Equation (2.9) is rarely used for design or data comparison, and is just mentioned for historical purposes.

Example 2.1

PCl_3 is an important impurity to remove in producing high quality polysilicon for use in solar arrays and electronic integrated circuits (like computer chips). Part of the purification process involves distillation, and so it is desirable to know the NBP of this compound with high confidence. Researching the NIST database shows an Antoine expression (Equation (2.8)), but the constants given are recalculated from a 1947 paper by Dan Stull published in I&EC. In making that paper's VP tables, he took data from several early-to-mid 20th century sources and plotted them on a Cox Chart (a graphical approximation method from the 1920's), then read "best fit" values from the Cox Chart at selected pressures. So this data source is questionable, with a possibility of data being "overly massaged". However from DeChema and Infotherm online global databases, experimental data is available from four more recent sources, with some mild disagreement in the pressure range of 1.0 atmospheres = 760 Torr = 101.325 kPa. It is decided to download the original data and develop an Antoine curve-fit, to best determine a most-likely NBP for PCl_3.

Table 2-1 shows the experimental data, sorted by temperature. DeChema and Infotherm report vapor pressures in kPa, so that pressure unit is used in the below calculation. To each data point row, columns for "1/T" and "Ln(P)/T" are added. The last two columns show the VP predicted by the Antoine regression and the difference between data point and predicted VP. Under the tabulated data, the results of the regression are shown, as well as the calculation of Antoine constants A, B, and C. And finally the best value for PCl3's NBP based on data along with an assessment of accuracy.

Table 2-1 Curve-fitting PCl3 VP data to the Antoine Equation, plus regression results

T, K	VP, kPa	Ln(VP), dependent variable	1/T, independent variable 1	Ln(VP)/T, independent variable 2	Antoine predicted VP, kPa	Prediction Difference
361.85	151.988	5.024	2.7636E-03	1.3884E-02	150.362	1.626
361.75	151.988	5.024	2.7643E-03	1.3887E-02	149.964	2.024
358.55	136.389	4.916	2.7890E-03	1.3709E-02	137.544	-1.155
355.75	126.790	4.843	2.8110E-03	1.3612E-02	127.191	-0.402
352.95	116.524	4.758	2.8333E-03	1.3481E-02	117.318	-0.794
349.15	104.525	4.649	2.8641E-03	1.3316E-02	104.680	-0.155
348.55	101.325	4.618	2.8690E-03	1.3250E-02	102.764	-1.439
348.45	101.325	4.618	2.8699E-03	1.3250E-02	102.447	-1.122
347.95	101.325	4.618	2.8740E-03	1.3273E-02	100.870	0.455
346.15	95.192	4.556	2.8889E-03	1.3162E-02	95.317	-0.125

T, K	VP, kPa	Ln(VP), dependent variable	1/T, independent variable 1	Ln(VP)/T, independent variable 2	Antoine predicted VP, kPa	Prediction Difference
343.75	88.259	4.480	2.9091E-03	1.3034E-02	88.213	0.046
343.15	86.526	4.460	2.9142E-03	1.2999E-02	86.490	0.036
338.15	73.461	4.297	2.9573E-03	1.2707E-02	72.952	0.508
337.55	72.127	4.278	2.9625E-03	1.2675E-02	71.425	0.703

The regression results using MS Excel's data analysis function show an R^2 = 0.9971 with 14 data points

Intercept = 9.7954; Variable 1 coefficient = -2640.04; Variable 2 coefficient =181.437

Calculate Antoine "A", "B", and "C" per the above procedure:

Intercept = 9.7954 = Antoine "A"

Variable 1 coeff. = -2640.04 so (- Intercept x Variable 2 coeff. - Variable 1 coeff.) = Antoine "B" = 862.795

Variable 2 coeff. = 181.437 = -1 x Antoine "C", so Antoine "C" = -181.437

The Antoine Equation for PCl_3 is determined as

$$\ln(P) = 9.7954 - \frac{862.795}{T - 181.437} \tag{2.12}$$

Solving Equation (2.12) for the atmospheric pressure of 101.325 kPa gives a T_b of 348.09°K = 74.94°C ⇒ 74.9°C .

The average absolute difference of (data -predicted) VP = the table's last column is 0.756 kPa

Note that if the NIST webbook values were used for "A", "B" and "C", a value of 348.34°K ⇒ 75.2°C would result. In this case, the NIST listed results were pretty close the experimental values' regression. Also note that a quick search on Wikipedia's website would have offered a T_b value of 76.1°C, or about a degree higher than actual results. But now that the best value of T_b is known, and there can also be a reasonable evaluation of value's accuracy based on the regression's average error. A 0.756 kPa error at 101.325 kPa pressure equates to a change in calculated T_b of 0.24°K, so the scientific answer to the question is 74.9± 0.2°C.

3: Critical Properties and Acentric Factor

Distillation Science (a blend of Chemistry and Chemical Engineering)

This is **Part III, Critical Properties and Acentric Factor** of a ten-part series of technical articles on Distillation Science, as is currently practiced on an industrial level. See also **Part I, Overview** for introductory comments, the scope of the article series, and nomenclature. The goal of this article is to explain how the critical properties and acentric factor are used, and how they may be estimated when reliable values are not available.

As discussed in **Part II**, the more basic vapor pressure equations have limitations that prevent them from being used in many modern industrial distillation systems, specifically at higher pressures. The concepts in this article will be used in **Part IV**, to introduce an improved vapor pressure equation, which removes the limitations of basic vapor pressure equations.

To implement the more sophisticated VP relationships and inter-compare the constants in **Part IV,** experimentally measured P vs T data needs to be put into reduced form. This is done only with the knowledge of the critical point, which is that singularity on the saturation line where the liquid and vapor phases become one. The critical temperature and pressure are identified as T_c and P_c. At the critical point, the specific molar volume is V_c.

$$T_r = T/T_c \tag{3.1}$$

$$P_r = P/P_c$$

$$V_r = V/V_c$$

(note that reduced properties have the advantage of being dimensionless)

Using reduced variables T_r, P_r and V_r from Equation (3.1) also allows introduction to the concept of the "Law of Corresponding States" (which is really not a scientific law per se, but more of a generally followed relationship). This "law" expresses the generalization that those properties dependent on intermolecular forces are related to the critical properties in the same way for all fluids. This concept underlies the development of several later articles of this series on Distillation Science, including **Parts IV** through **VII**.

The critical compressibility is Z_c, as defined as:

$$Z_c = \frac{P_c V_c}{R T_c} \tag{3.2}$$

which is also dimensionless.

Two other properties are important to note:

- T_b is defined as the atmospheric pressure boiling point of a fluid, with T_{br} as the reduced value of the atmospheric boiling point
- ω is the acentric factor, which is a measure of molecular complexity as defined as:

$$\omega = -\log_{10}(VP) - 1 \tag{3.3}$$

where the VP (in atmospheres) is evaluated at T_r= 0.7

The term “acentric factor” comes from Kenneth Pfizer's observation in 1955 that compact and near-spherical molecules have close to zero values when their ω is calculated. For example, neon, argon and krypton have ω values of -0.04, 0.00 and 0.00, respectively (helium has a more slightly negative ω, the value depending on isotope). Methane (CH_4) has a ω of 0.011; and the molecularly larger silane (SiH_4) has an ω of 0.099. An even larger and more complex molecule, like carbon tetrachloride (CCl_4) has an ω of 0.193; whereas silicon tetrachloride ($SiCl_4$) has an ω of 0.248. So valuation of the acentric factor from VP data and knowledge of the critical point is a good validation check against the known molecular size and shape.

For Equations of State (**see Part V**) that are more advanced than Van der Waals, the acentric factor is a required property, along with critical temperature and pressure. It is also used along with critical properties in the estimation of binary interaction parameters in **Part VII**.

As mentioned in **Part II**, the basic VP relationships have acceptable accuracy to obtain good values of T_b from available data. Experimental data is often taken near atmospheric pressure, but not exactly at one atmosphere absolute (i.e, 760 mmHg = 760 Torr = 101.325 Pa). Instead of blindly accepting a value of T_b from a handbook or single website, the practicing scientist or engineer should consider the data’s source and reliability. There is ready internet availability of atmospheric boiling points, critical property data, and acentric factors, especially on the NIST WebBook and global websites like Dechema, Infotherm (recently acquired by John Wiley and Sons from FIZ Chemie Berlin).

A recommended way to sort through the dizzying amount of critical property data (i.e., T_c , P_c , and V_c) is to use the correlation techniques of Lyderson (*"Estimation of Critical Properties of Organic Compounds", University of Wisconsin College of Engineering, 1955)* and others, which are specifically geared toward organic compounds. With some algebraic manipulation and adjustment of Lyderson's parameters (since the usage intent here is for polar compounds that are not organic, but are not ionic either), and based on data across homologues, the following relationships seem to hold and are recommended for both validating questionable data or filling in "holes" of no data:

For T_b, the general correlating relationship within a homologue (e.g., from silane to silicon tetrachloride, or from phosphine to trichlorophosphine) :

$$(T_b \times MW)^n = A + B \times MW \tag{3.4}$$

where MW is the molecular weight. For chlorosilanes, chloromethanes, chlorophosphines, etc, this relationship shows a best fit with an exponent "n" of 0.8320. The constants A and B for the homologue are typically set by the hydride and chloride, since that data is usually more reliable. Where the core atom is not just carbon or silicon, but a blend (i.e, a methyl silane), an exponent of 0.8455 fits the data a bit better. This contrasts with organic compounds, whose data tends to fit best with somewhat lower exponents that are closer to 0.80.

For T_c (critical temperature), the best correlation within a homologue is found to be:

$$\frac{T_c - T_b}{T_c} = A + B \times MW \tag{3.5}$$

where MW is the molecular weight and the constants A and B for the homologue are typically set by the hydride and chloride, as long as there is no organic content and the fluids are monomeric.

For P_c (critical temperature), use

$$(\frac{MW}{P_c})^n = A + B \times MW \tag{3.6}$$

where MW is the molecular weight, exponent "n" = 0.5672, and constants A and B for the homologue are typically set by the hydride and chloride (as long as there is no organic content and the fluids are monomeric). For a pure organic compound, the exponent "n" should be Lyderson's suggested 0.5000; for methyl silanes (e.g, dimethyl silane) the best exponent is 0.4550; and for Group III (dimeric bridge-bonded) compounds the best exponent fit is 1.03-1.08 (1.03 for di-gallanes and 1.08 for diboranes).

For V_c (critical molar volume), Lyderson's rule of

$$V_c = A + B \times MW \tag{3.7}$$

is probably still the best, where MW is the molecular weight and constants A and B for the homologue are typically set by the hydride and chloride. For some homologues, there is a possibility that the Vc vs MW relationship is not exactly linear, but rather has a slight concave quadratic nature; but the data is rarely good enough to determine such. Of all the critical properties, V_c is the hardest to experimentally measure, and is frequently estimated.

There is a way to get around some of the uncertainty with V_c, and that is to validate (or make minor V_c adjustments) based on the pattern of Z_c values in the homologue where

$$Z_c = \frac{P_c \times V_c}{R \times T_c} \tag{3.8}$$

Within each homologue there is a characteristic "check-mark" pattern to Z_c values, when plotted against the number of hydrogen to chlorine swaps of the homologue (i.e., the homologue's hydride has zero swaps, the monochloride one swap, the dichloride two swaps, etc). See below example data plot **Figure 3-1** for the chlorosilane and chlorophosphine homologues.

The homologue's hydride (which tends to be a fairly spherical shaped molecule with minimal dipole moment) typically has a Z_c in the range of 0.27-0.29. The first swap replaces a small "H" atom with a larger "Cl" atom, and re-orients the molecular shape to have a significant dipole, dropping the Z_c value by about 10%. Then the next swaps reduce the dipole, until the molecular shape returns closer to spherical, although significantly larger in diameter. Typically the Z_c of the completely chlorinated homologue fluid has a slightly larger Z_c than the homologue's hydride.

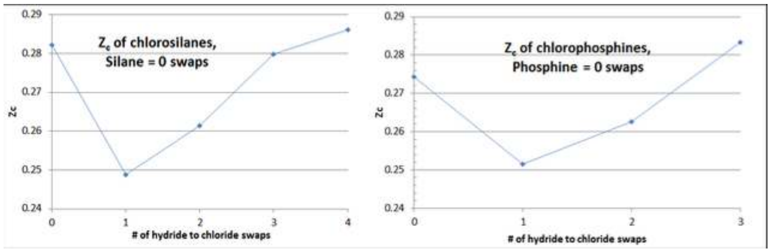


Figure 3-1, Comparison of Z_c pattern within two homologues

To the extent that this "checkmark" shape is not observed in plotting calculated Z_c values per Equation 3-8, the "culprit" is almost always the value of V_c, which allows for some correction. The value of Z_c is known to have a significant relationship to molecular shape and complexity, although the stronger relationship between Z_c and the acentric factor, ω, is not so easily generalized as in organic compounds.

The importance of accurate values of T_b , T_c , P_c (and ω to an extent) is in calculating vapor pressure using the advanced relationship of **Part IV**. Having an accurate value of ω is more important, along with T_c, P_c in calculating the Equation of State for a fluid, in **Part V**. Having good values for ω and V_c are important to the estimation of binary interaction parameters in **Part VII**.

When two different types of swaps are made in a slightly polar molecule, an alternate technique is used to best correlate fluid properties (T_b , T_c , P_c , V_c , and ω), with a good example being methyl chlorosilanes. With methyl chlorosilanes, one or more methyl groups are substituted for hydrogen atoms (i.e., Si-H swapped to Si-CH_3), with the remaining Si-H bonding possibly swapped to Si-Cl. Another example would be swapping C-H for C-Cl and C-F bonding in chloro-fluoro methanes. There is no concise formula that governs such a dual substitution. Instead the concept of neural networks is used.

In this technique, the two base curves are laid out from the same starting compound (silane in the example below), and on each base curve the pertinent physical property values shown for each substituted homologue (chlorosilanes and methyl silanes being the base curves in the example below) vs molecular weight. For each amount of combined substitution (100%, 75% and 50% substitution = 0%, 25%, and 50% Si-H remaining) the data points are connected to their respective base curves. For example, on the 75% substitution amount, one would curve-connect trichlorosilane, methyl dichlorosilane, dimethyl monochlorosilane, and trimethyl silane.

Additional curve-connections could be made for values of constant amount of chlorine or methyl substitution (e.g., curve-connecting monochlorosilane, methyl monochlorosilane, dimethyl monochlorosilane, and trimethyl monochlorosilane). Using this method, the data on the dual-substituted fluids can be validated and adjustments made so that the curve-connections are smooth.

In the example in below Figure 3-2, the values of P_c for methyl monochlorosilane (*sic*, me-mono) and dimethyl monochlorosilane (*sic*, di-me mono) are hardened up from rough measurements.

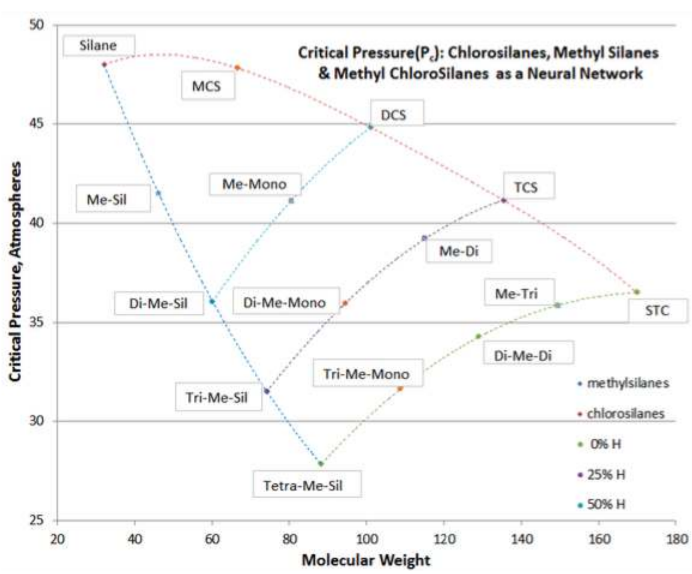


Figure 3-2, Neural Network for methyl chlorosilanes' P_c **values**

The value of the neural network technique is to make adjustments where data is poor or the estimation is of questionable accuracy. The presumption of neural networks is that for a class of compound, the continuum of structural changes should be uniform and internally consistent (although not necessarily linear).

Neural networks were constructed for all the pertinent properties of methyl chlorosilanes, and the technique found to be rather simple to deal with graphically using MS Excel. The resulting values were included in Table 3-3 below, for chlorosilanes and their common impurities, whose properties were validated using the correlating techniques of Equation (3.4) through Equation (3.8).

Table 3-3, Key properties, arranged by Group and Homologue

Fluid	MW	T_b	T_c	P_c	V_c	Z_c	ω
Group III(A)							
B_2H_6	27.670	180.54	289.70	39.58	173.10	0.2882	0.1254
B_2H_5Cl	62.119‡	216.37	346.33	37.13	189.92	0.2481	0.1283
BH_2Cl as monomer	42.284	237.72	379.56	37.10	206.73	0.2463	0.1315
$BHCl_2$ as monomer	82.722	266.07	422.71	37.63	240.37	0.2608	0.1389
B_2HCl_5 hypothetic	199.893‡	276.67	438.47	37.92	257.18	0.2711	0.1433
BCl_3	117.169	285.88	451.95	38.20	274.00	0.2822	0.1468
$AlCl_3$ as monomer	133.341	466.86*	625.70	26.00	261.80	0.1326	0.3474
$Ga_2H_4Cl_2$ as dimer	214.383‡	350.86	545.93	40.94	231.9	0.2120	0.2726
$Ga_2H_2Cl_4$ as dimer	283.273‡	421.79	639.69	41.02	247.5	0.1943	0.3525
Ga_2Cl_6 as dimer	352.162‡	473.49	694.00	37.70	263.0	0.1741	0.4504
Group IV(A)							
SiH_4	32.117	161.75	269.65	47.99	130.07	0.2821	0.09860
SiH_3Cl	66.562	242.75	396.65	47.82	169.32	0.2488	0.1252
SiH_2Cl_2	101.007	281.45	449.45	44.83	215.05	0.2614	0.1589
$SiHCl_3$	135.452	306.15	479.15	41.15	267.28	0.2797	0.2090
$SiCl_4$	169.896	330.72	506.95	36.50	326.00	0.2860	0.2482
$SiH_3(CH_3)$	46.144	216.48	348.35	41.53	185.20	0.2691	0.1264
$SiH_2Cl(CH_3)$	80.589	278.82	439.11	41.16	233.90	0.2672	0.1793
$SiHCl_2(CH_3)$	115.034	314.17	489.18	39.25	287.81	0.2814	0.2269
$SiCl_3(CH_3)$	149.479	339.72	517.61	35.86	345.70	0.2919	0.2655
$SiH_2(CH_3)_2$	60.169	252.86	399.17	36.05	245.84	0.2706	0.1604
$SiHCl(CH_3)_2$	94.615	305.30	471.35	35.94	300.32	0.2791	0.2264
$SiCl_2(CH_3)_2$	129.061	342.89	519.21	34.26	358.20	0.2880	0.2740

Fluid	MW	T_b	T_c	P_c	V_c	Z_c	ω
GeH_4	76.642	184.93	307.98	54.77	128.28	0.2780	0.1270
GeH_3Cl	111.087	302.91	495.10	49.95	180.17	0.2215	0.1476
GeH_2Cl_2	145.532	334.60	536.93	45.36	236.71	0.2437	0.1721
$GeHCl_3$	179.976	343.54	541.41	41.42	283.96	0.2647	0.2012
$GeCl_4$	214.421	357.28	553.16	38.10	335.86	0.2819	0.2334
SnH_4	122.742	221.07	360.20	51.70	152.90	0.2674	0.1619
$SnCl_4$	260.521	387.21	591.85	36.95	351.20	0.2672	0.2625
Group V(A)							
PH_3	33.998	185.41	324.75	64.51	113.33	0.2743	0.03052
PH_2Cl	68.443	273.09	463.73	62.29	153.63	0.2515	0.0750
$PHCl_2$	102.888	318.44	524.74	55.82	202.52	0.2625	0.1362
PCl_3	137.333	349.25	558.95	50.00	260.00	0.2834	0.2117
$POCl_3$	153.331	379.00	605.21	47.59	276.00	0.2645	0.1993

Table 3-3, Continued

AsH_3	77.945	210.73	373.00	65.12	132.50	0.2819	0.01341
AsH_2Cl	112.390	300.41	515.99	64.33	174.85	0.2657	0.0573
$AsHCl_2$	146.835	359.66	600.00	61.57	218.29	0.2730	0.1153
$AsCl_3$	181.281	403.30	654.00	58.35	259.56	0.2822	0.1875
SbH_3	124.781	256.09	446.20	66.61	157.20	0.2860	0.01659
$SbCl_3$	228.115	794.05	794.05	68.85	268.00	0.2832	0.3309
n-pentane	72.149	309.16	470.05	32.86	310.0	0.2641	0.2393
iso-pentane	72.149	300.82	460.56	33.17	307.1	.2695	0.2147

In above Table 3-3, those entries that are marked with a ǂ have their molecular weight given as the dimer. For $AlCl_3$, the asterisk on the T_b entry is for the best correlating value for the **Part IV** VP equation, even though it is below the triple point. Group IIIA compounds can feature “bridge-bonding” and can be either monomeric or dimeric. The two pentane entries at the end of the table are included because of their significance as electronic impurities, and to show that the techniques can be extended to organics.

With Group IIIA, some compounds are not included in a homologue simply because they are impossible structurally or are unstable. An example would be $B_2H_3Cl_3$, which would put too much strain on the B-B bridge-bond. Another is AlH_3 which cannot exist as a stand-alone liquid with vapor pressure at normal processing conditions, but tends to form stable hydrides with Group I metals such as $LiAlH_4$. However, B_2HCl_5 is included (but noted as hypothetic), since some researchers claim to have seen its presence in low levels of TCS.

4: New Vapor Pressure Equation

Distillation Science (a blend of Chemistry and Chemical Engineering)

This is **Part IV, New Vapor Pressure Equation** of a ten-part series of technical articles on Distillation Science, as is currently practiced on an industrial level. See **Part I, Overview** for introductory comments, the scope of the article series, and nomenclature. **Parts II** and **III** are pre-requisite to **Part IV**.

Previous **Part II, Vapor Pressure** deals with the pure component equations normally found in textbooks, but which have limitations in industrial application. **Part III, Critical Properties and Acentric Factor** develops those parameters used to inter-relate conventional units of temperature, pressure and volume to their reduced property equivalents, as well as the Pfizer acentric factor that is used in modern Equations of State.

Several thermodynamically consistent vapor pressure (VP) equation forms have been developed in the past purposely for organic fluids, and typically have the stipulation that they do not work with either polar or heavily hydrogen-bonded fluids. Being both slightly polar in nature as well as hydrogen-bonded, chlorosilanes and their impurities fall exactly in this exclusion, and therefore make good continuing examples.

In this article, temperature and pressure are used in their reduced form, since this allows better inter-comparison between fluids, is dimensionless, and allows a more global future development. The reduced form is obtained by simply dividing conventional temperature, T, by the fluid's critical temperature=> $T_r = T/T_c$. Similarly, reduced pressure is conventional pressure divided by the fluid's critical pressure => $P_r = P/P_c$. Temperature and pressure must be in absolute units (e.g., ºK not ºC)

Referring back to Part II, Equation 2-4, the Clausius-Clapeyron equation is given in reduced format, using ψ as notation for the thermodynamic-based derivative of the natural log of (reduced) vapor pressure with respect to the inverse of (reduced) absolute temperature:

$$\psi = \frac{-d(\ln P_r)}{d(1/T_r)} = \frac{\Delta H_v}{\Delta Z R T_c} \tag{4.1}$$

In order to have an improved VP equation form that can be used for the wide range between T_b and T_c (i.e., from atmospheric pressure to the critical point), that relationship should meet the following criteria:

1. The equation form should exactly have a reduced VP value of $1/P_c$ at T_b/T_c (i.e., yields the known atmospheric boiling point at atmospheric pressure), which is a well-measured lab value and is readily estimated based on structural considerations (see **Part III** article).
2. The equation form should exactly have a reduced VP value of unity at the critical temperature (i.e., match the measured value of the critical point), where the liquid meniscus disappears. This property can also be readily estimated based on structural considerations (see **Part III** article).
3. Give a reasonable value for reduced VP at T_r=0.7, corresponding to the definition of the Pfizer acentric factor, which has a basis in molecular complexity (see **Part III** article).
4. Meet the two thermodynamic consistency Reidel tests, as the critical point is approached (that "α" monotonically drop to its lowest value at critical, and that the temperature derivative of α = zero at critical). See the left-most part of Equation (4.4) for the definition of "α".
5. It must be noted that while this new vapor pressure relationship exactly matches the known atmospheric pressure boiling point, its differential form will not yield the known atmospheric pressure latent heat of vaporization

.Such a VP equation form would allow evaluation of modern Equations of State (see **Part V** article), Fugacity considerations (see **Part VI** article), and Binary Interaction Parameters (see **Part VII** article). Preferably the VP equation form would reasonably fit most fluids (both naturally occurring as well as synthetic), including polar compounds and those with substantial hydrogen bonding. Note that the VP equations given in **Part II** work over only narrow ranges; however above criteria #1 and #2 require the VP equation form to span a very broad range. In the text "The Properties of Gases and Liquids", by Prausnitz, et al, several alternate VP equation forms are discussed, all of which meet the above four criteria.

In his doctoral thesis at Syracuse University in 1965, under the guidance of Leonard Stiel, Richard Thek proposed exactly such an equation form to fill the technology gap. It was subsequently published by the AIChE in 1966. Unfortunately the computing power at that time was very limited, so equation forms with iterative solutions could not be used; and the internet did not exist to allow ready access to global data taken over the last many decades. Using modern computer processing speed and programming, these limitations have been resolved, and the constants determined for all chlorosilanes and their electronic impurities.

As given by Equation (4.2) below, the original Thek-Stiel VP equation in its differential form has two terms: the first that governs at lower pressures close to atmospheric, and the second that takes over as the critical point is approached. While Thek set variables “q” and “k” equal to constants in order to avoid iterative solutions, with modern programming these can now be evaluated as true variables. The derivation of the final Modified Thek-Stiel VP Equation is available on request, but is not reproduced in these articles for the sake of brevity.

$$\frac{d(\ln P_r)}{d(1/T_r)} = \frac{A}{T_r^2}[1 - B_1 T_r + B_2 T_r^2 - B_3 T_r^3 + B_4 T_r^4 ...] + \frac{c}{T_r^2}(T_r^n - k) \tag{4.2}$$

The several “B” constants result from the binomial expansion of the general Watson relationship for ΔH_v. After dropping the B_4 and higher expansion terms as non-significant mathematically and integrating, the resulting reduced vapor pressure equation is:

$$\ln(P_r) = A[B_0 - T_r^{-1} - B_1 \ln T_r + B_2 T_r - (1/2) B_3 T_r^2] + c\left[\frac{T_r^{n-1} - 1}{n-1} + k(T_r^{-1} - 1)\right] \tag{4.3}$$

- $A = \dfrac{\Delta H_{vb}}{RT_c(1 - T_{br})^q}$
- $B_0 = 1 - B_2 + (\frac{1}{2})B_3$
- $B_1 = q$
- $B_2 = \dfrac{q(q-1)}{2!} = \dfrac{q(q-1)}{2}$
- $B_3 = \dfrac{q(q-1)(q-2)}{3!} = \dfrac{q(q-1)(q-2)}{6}$
- $c = \dfrac{\alpha_c - A(1 - B_1 + B_2 - B_3)}{1-k}$
- $n = (1-k) + \dfrac{A}{c}(1 - B_2 + 2B_3)$

The valuation of equation parameters ΔH_{vb}, q, k, and α_c are done iteratively using the techniques in **Part X, Convergence Strategy**, based on vapor pressure data, the molecule’s structure, and the value of T_{br} (the reduced atmospheric boiling point, $T_b/T_c = T_{br}$). P_r must be unity at the critical point ($T_r = 1$), which allows valuing the integration constant of Equation 4-3. The two important thermodynamic derivative functions are:

$$\frac{d(\ln P_r)}{d(\ln T_r)} = \alpha = A(T_r^{-1} - B_1 + B_2 T_r - B_3 T_r^2) + c(T_r^{n-1} - \frac{k}{T_r}) \tag{4.4}$$

$$\frac{-d(\ln P_r)}{d(1/T_r)} = \frac{\Delta H_v}{\Delta Z R T_c} = \psi = A(1 - B_1 T_r + B_2 T_r^2 - B_3 T_r^3) + c(T_r^n - k) \tag{4.5}$$

The "A" and "B" constants of Equations (4.2), (4.3), and (4.4) are the same as Equation (4.5). The evaluation of the Reidel derivative function "α" in Equation (4.4) is used to establish thermodynamic consistency, and allows valuing an equation parameter. Note that the Clapeyron derivative of Equation (4.5), ψ, is mathematically identical to that in Equation (4.1). ψ will be used **Part V** to establish the latent heat, ΔH_v, as well as the saturated vapor and liquid density of the fluids at the various conditions in the distillation column modeling.

Using the results of **Part III**, the Modified Thek-Stiel VP Equation constants are given below in Table 4-1, for chlorosilanes and electronic impurities found in the manufacture of high purity silicon. For brevity, the values of acentric factor, ω, and the critical Riedel derivative, α_c, are not given in the table. They can be calculated from the other equation constants. In some instances, an electronic impurity can have either monomeric or dimeric form, but the form is noted in the table. Where a compound is not stable, the reader will note a “hole” in the table (e.g., AlH_3 and Ga_2H_6 are not stable as liquids or vapors).

Because the solution to the Modified Thek-Stiel VP Equation is necessarily iterative, the application of this VP relationship requires the use of some modern computing tools to perform such iterative calculations (aka “nested loops”). Depending on preference, this could be done as a macro in a spreadsheet program like MS Excel, or a stand-alone program developed in BASIC (or MatLab or Fortran). As mentioned above, some tips on convergence strategy are given in **Part X**.

Table 4-1 is rather expansive, to illustrate how general the new VP relationship is. Table organization is primarily done by Periodic Table Group (e.g., Group III, Group IV, and Group V) of the molecule’s core atom, and then by the homologue from hydride to chloride. Methyl chlorosilanes are included as examples of good application to fluids that form the border between classically organic and inorganic. Two of the pentanes are included, which happen to be of industrial concern as electronic impurities, to illustrate that the new VP equation is useful for organic fluids as well.

While volatile Group II, Group VI and Group VII fluids seem to follow the new VP relationship as well, there are no entries given in Table 4-1 since these fluids are generally not of concern in the production of electronic materials. The reader is encouraged to follow the evaluation techniques and stratagems given in these articles to further develop use of the new VP equations.

Table 4-1, VP solution constants, by Group and Homologue

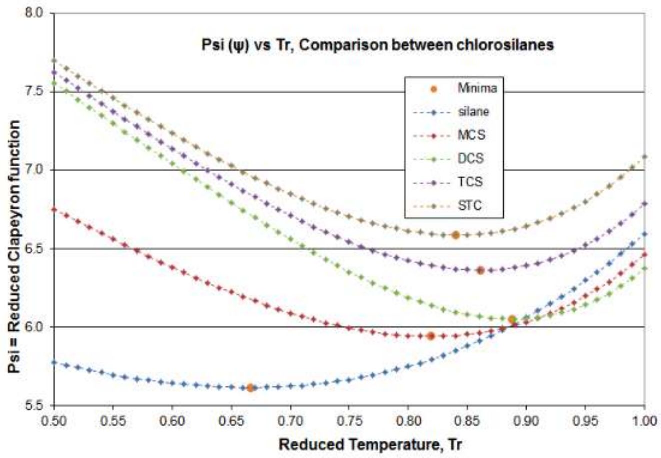
Fluid	MW	A	B_0	B_1	B_2	B_3	c	n	k
Group III(A)									
B_2H_6	27.670	8.9910	1.1494	0.37848	-0.11762	0.063573	2.8701	4.7796	0.1197
B_2H_5Cl	62.119‡	8.1915	1.1494	0.37867	-0.11764	0.063578	3.4560	3.8431	0.1073
BH_2Cl as monomer	42.284	8.3872	1.1495	0.37888	-0.11766	0.063583	3.2767	4.0925	0.0938
$BHCl_2$ as monomer	82.722	8.9910	1.1495	0.37937	-0.11772	0.063596	2.7577	4.9953	0.0636
BCl_3	117.169	9.5652	1.1496	0.37988	-0.11779	0.063609	2.2357	6.2975	0.0291
$AlCl_3$ as monomer	133.341	11.9183	1.1512	0.39300	-0.11928	0.063892	2.2700	7.5185	0.0291
$Ga_2H_4Cl_2$ as dimer	214.383‡	7.0714	1.1506	0.38810	-0.11874	0.063799	5.3346	2.5583	0.0938
$Ga_2H_2Cl_4$ as dimer	283.273‡	10.8499	1.1513	0.39333	-0.11931	0.063898	3.3656	4.9567	0.0636
Ga_2Cl_6 as dimer	352.162‡	11.1760	1.1520	0.39974	0.11997	0.063996	3.8566	4.5874	0.0291
Group IV(A)									
SiH_4	32.117	7.4652	1.1492	0.37673	-0.11740	0.063525	3.6247	3.4716	0.0914
SiH_3Cl	66.562	8.8830	1.1494	0.37847	-0.11761	0.063572	2.7614	4.9286	0.0756
SiH_2Cl_2	101.007	9.9221	1.1497	0.38068	-0.11788	0.063629	2.1589	6.6627	0.0598
$SiHCl_3$	135.452	9.9796	1.1501	0.38395	-0.11827	0.063708	2.5687	5.7950	0.0446
$SiCl_4$	169.896	10.0240	1.1504	0.38651	-0.11856	0.063765	2.8465	5.3590	0.0291
$SiH_3(CH_3)$	46.144	9.0535	1.1494	0.37855	-0.11762	0.063574	2.6380	5.1963	0.0758
$SiH_2Cl(CH_3)$	80.589	10.1402	1.1499	0.38201	-0.11804	0.063662	2.1796	6.7338	0.0600
$SiHCl_2(CH_3)$	115.034	9.1805	1.1503	0.38512	-0.11840	0.063735	3.3346	4.3854	0.0446
$SiCl_3(CH_3)$	149.479	10.1187	1.1506	0.38512	-0.11840	0.063789	2.9357	5.2665	0.0291
$SiH_2(CH_3)_2$	60.169	8.2747	1.1497	0.38077	-0.11789	0.063631	3.4554	3.9214	0.0603

$SiHCl(CH_3)_2$	94.615	9.1914	1.1503	0.38509	-0.11840	0.063734	3.3232	4.4012	0.0447
$SiCl_2(CH_3)_2$	129.061	10.1007	1.1507	0.38820	-0.11875	0.063801	3.0279	5.1285	0.0291
GeH_4	76.642	8.1986	1.1494	0.37858	-0.11763	0.063575	3.3615	3.9445	0.0914
GeH_3Cl	111.087	8.6130	1.1496	0.37993	-0.11779	0.063610	3.1743	4.3025	0.0756
GeH_2Cl_2	145.532	8.8966	1.1498	0.38154	-0.11798	0.063650	3.1184	4.4929	0.0598
$GeHCl_3$	179.976	9.1230	1.1501	0.38344	-0.11821	0.063696	3.1490	4.5641	0.0446
$GeCl_4$	214.421	9.4041	1.1503	0.38554	-0.11845	0.063744	3.1654	4.6724	0.0291
SnH_4	122.742	9.0298	1.1497	0.38087	-0.11790	0.063634	3.0671	4.5745	0.0914
$SnCl_4$	260.521	10.1994	1.1506	0.38745	-0.11867	0.063744	2.8471	5.4354	0.0291
Group V(A)									
PH_3	33.998	7.8041	1.1485	0.37228	-0.11684	0.063396	2.8050	4.3644	0.1009
PH_2Cl	68.443	8.6412	1.1490	0.37518	-0.11721	0.063482	2.4920	5.2267	0.0766
$PHCl_2$	102.888	8.5412	1.1495	0.37919	-0.11770	0.063591	3.0719	4.3813	0.0527
PCl_3	137.333	7.2366	1.1501	0.38413	-0.11829	0.063712	4.4289	2.9789	0.0291
$POCl_3$	153.331	9.7746	1.1500	0.38331	-0.11819	0.063693	2.4586	5.9576	0

Table 4-2, VP solutions, by Group and Homologue, continued

Fluid	MW	A	B_0	B_1	B_2	B_3	c	n	k
Group V(A),cont'd									
AsH_3	77.945	7.4208	1.1484	0.37116	-0.11670	0.063362	2.9474	4.0298	0.1009
AsH_2Cl	112.390	8.5242	1.1488	0.37403	-0.11707	0.063448	2.3972	5.3468	0.0766
$AsHCl_2$	146.835	9.3548	1.1493	0.37782	-0.11754	0.063555	2.1821	6.2831	0.0527
$AsCl_3$	181.281	10.0281	1.1499	0.38254	-0.11810	0.063675	2.2519	6.5171	0.0291
SbH_3	124.781	8.5409	1.1484	0.37136	-0.11673	0.063368	2.0491	6.0821	0.1009
$SbCl_3$	228.115	7.1200	1.1511	0.39192	-0.11916	0.063873	5.3455	2.6317	0.0291
n-pentane	72.149	10.0508	1.1504	0.38593	-0.11849	0.063753	2.6269	5.7673	0
iso-pentane	72.149	9.9797	1.1502	0.38432	-0.11831	0.063717	2.4514	6.0715	0

Two plots are offered in Figure 4-3 to validate the principles of this Distillation Science article. These plots are for the ψ(Psi) function of Equation (4.5) and the α(alpha) function of Equation (4.4), for the chlorosilane homologue. The derivative function ϕ goes through the required minimum, which produces the expected inflection in the vapor pressure curve, typically around T_r= 0.80-0.86. Silane's minimum ϕoccurs at T_r=0.67 and DCS's minimum ϕ occurs at T_r= 0.89. The fact that two fluids of this homologue are outliers illustrates why chlorosilane vapor pressures are so poorly fit by VP expressions that are intended for use with hydrocarbons. However, it can be seen how the Reidel α function does fit the thermodynamic consistency requirements: for all homologue fluids, the α function asymptotes to a constant value (α_c) as the critical point is approached, with the slope of the α function going to zero as critical is approached.



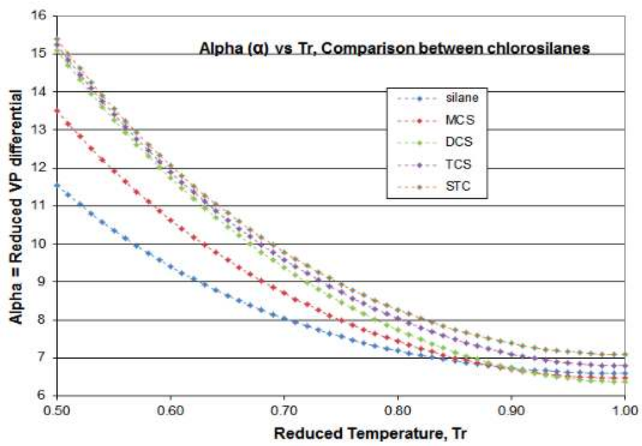


Figure 4-3 Derivative functions ψ **and** α **, for chlorosilanes**

Another validation check is shown in Figure 4-4, comparing the vapor pressures of chlorosilanes and chlorophosphines, which are distillation separation applications in all commercial chlorosilane purification.

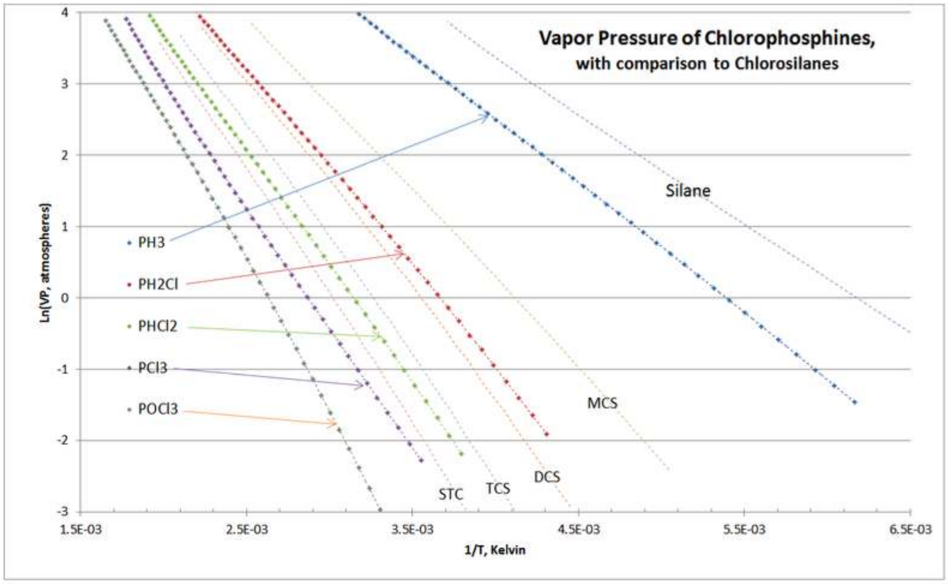


Figure 4-4, Cross-Plot of Chlorophosphines vs Chlorosilanes

As expected, the chlorophosphine homologue's VP curves "nest" around those of the chlorosilanes. There are no cross-overs of the curves, and the location of chlorophosphine vapor pressure curves lie exactly as they are experienced in commercial practice of chlorosilane purification distillation columns.

Since the math in Equation (4.3) is somewhat involved, an example is provided with calculations.

Example: At 3.50 atmospheres pressure what is the boiling point of trichlorosilane (TCS)?

From **Part III**, Table 3-3, the pertinent property values for TCS are: T_c=479.15°K and P_c=41.15 atm. MW, T_b , V_c and Z_c are not needed for this example. From **Part IV**, the VP solution constants for TCS are: A=9.9796; B_0=1.1501; B_1=0.38395; B_2=-0.11827; B_3=0.063708; c=2.5687; n=5.7950; and k=0.0446.

As all advanced VP equations are, determining the TCS boiling point at 3.50 atmospheres is iterative; so it is nice to have a VP plot to get a good first guess. Using above Figure 4-4, for

\ln (VP) =\ln (3.50) = 1.2528, it looks like 1/T= 2.89E-3 might be a good first guess => T=346°K; so first guess T_r=346/479.15 =0.7221. Plugging that first guess value of T_r into the T-S VP equation with the constants above:

$\ln(P_r) = A[B_0 - T_r^{-1} - B_1 \ln T_r + B_2 T_r - (1/2) B_3 T_r^2] + c[\frac{T_r^{n-1} - 1}{n-1} + k(T_r^{-1} - 1)]$ yields a reduced VP value of P_r = 0.08273; so VP=0.08273*41.15 = 3.404 atm, which is just a bit less than the desired 3.50 atm. So T needs to increase a little from the 346°K initial first guess.

Re-doing the TCS VP equation with T= 347°K yields a VP of 3.50 atm (note: 347.05°K is the answer, if the solution is worked out to two decimal places of temperature). So the boiling temperature at 3.50 atmospheres is 347°K or 73.9°C.

5: Equation of State

Distillation Science (a blend of Chemistry and Chemical Engineering)

This is **Part V, Equation of State** of a ten-part series of technical articles on Distillation Science, as is currently practiced on an industrial level. See also **Part I, Overview** for introductory comments, the scope of the article series, and nomenclature. **Part V, Equation of State** explains why an Equation of State (EOS) is needed to evaluate the pure-component physical properties that are used in distillation science, reviews the various options and makes a recommendation. Also included are techniques for solving the EOS cubic equations via spreadsheet (i.e., MS Excel) and some work-arounds near the critical point. This article uses the critical constants and acentric factor of **Part III**. The EOS solutions of this article will feed values to **Part VI, Fugacity**.

The general PVT relationship for a fluid is

$$PV = ZRT \tag{5.1}$$

where the fluid could be a sub-cooled liquid, a saturated liquid (i.e, at its boiling point), a saturated vapor (i.e., at its dewpoint), or a superheated vapor (i.e., a gas). In the ideal situation (ambient pressure and temperature), Z_L is very close to zero for either a sub-cooled or saturated liquid, and Z_v is close to unity for either a saturated or superheated vapor. The Ideal Gas Law has $Z_V = 1$, so $\Delta Z = (Z_v - Z_L) \approx 1$

ΔZ is an important parameter to Equation 2-2 and 4-1 in previous articles **Part II** and **Part IV**. For calculations in this and other Parts, temperature and pressure are in absolute units (°K and atmospheres), and volume in molar units (e.g., cc/mole). In that case, the gas constant R has units of 82.057 atm-cc/mole-K.

Van der Waals EOS and more recent improvements

For a real fluid, Z is never 0 and almost never =1, but rather a value between 0 and 1, except for gases at high temperatures and pressures (where Z_v >1). While the Ideal Gas Law is a nice introductory concept, it is rarely used in distillation science. Instead, a more accurate expression is needed, and so Equation (5.1) is re-written as

$$Z = \frac{PV}{RT} \tag{5.2}$$

and different models are used to give results: Z as a function of T & P. The oldest is the Van der Waals EOS from the 1873 based on a model that molecules are hard bodies that take up space and have inter-particle forces. Note that an Equation of State covers all non-solid phases, so for saturated systems (i.e., fluids at their boiling/condensation point) it will have two real solutions: one for the vapor and one for the liquid: Z_V and Z_L. For a non-saturated system (i.e., a super-heated vapor or sub-cooled liquid) it will have only one real solution.

Written in terms of Z, the Van der Waals EOS is

$$Z = \left(P + \frac{a}{V^2}\right) \times (V - b)/RT \tag{5.3}$$

where a and b are constants derived from just the fluid's critical properties. While a good general approximation at lower pressures (better than the Ideal Gas Law), this expression becomes less accurate as the critical point is approached, and it always yields 0.375 as Z_c. There are very few fluids that have such a high Z_c - most are closer to 0.30. Solutions for saturated phase values, Z_v and Z_L, require manipulation of a cubic equation and solving its roots. However inaccurate it may be, it does allow estimation of the saturated vapor and liquid molar densities: $\rho_\nu = 1/V_\nu$ and $\rho_L = 1/V_L$, as well as $\Delta Z = Z_v - Z_L$.

In **Part IV**, Equation 4-1 showed how the latent heat of vaporization, ΔH_v, is calculated from the differential Clausius-Clapeyron equation, once a good vapor pressure relationship is known and ΔZ is closely determined. And in **Part VI**, it will be seen how fugacity coefficients are calculated from an EOS, to partially account for some departures to ideal behavior in mixtures of fluids.

In 1949, the Redlich-Kwong EOS improved on Van der Waals, and in 1972 Soave made a modification to yield the S-R-K EOS, by adding in the effect of the acentric factor, ω. Further improvement was done in a general-application EOS by Peng-Robinson in 1976. Since then, there have been many additional EOS proposed for specialized applications which give better results, but require additional data the additional constants. A full history of various EOS's is given in Wikipedia.

For purposes of practical chlorosilane distillation design of fugacity (see **Part VI**), the Peng-Robinson (P-R) EOS appears to give satisfactory results. P-R does not appear to be accurate enough to extend to closely estimating saturated phase densities of chlorosilanes, or latent heats. If enough data were available, it might be possible to determine which of the newer, more specific EOS forms is accurate enough for those derivative properties (i.e., superior to Peng-Robinson). Unfortunately, that data does not yet exist (although a few key data points show the failings of P-R).

Regardless of the fluid parameters chosen, the P-R EOS yields a Z_c of 0.307 at the critical point. While this Z_c value is still higher than that of chlorosilanes, the accuracy of Z_v and Z_L values are adequate **for fugacity estimation**, through 90% of critical pressure (P_c). Since commercial applications only rarely approach 90% of P_c for reasons of process and equipment design, the P-R EOS is an acceptable compromise. It must be pointed out that the P-R EOS should not be used in any calculations that involve differential or integral calculus manipulation of Z, since erroneous (or impossible) values result.

Peng-Robinson Equation of State

$$p=\frac{RT}{V_m-b}-\frac{a\alpha}{V_m^2+2bV_m-b^2} \tag{5.4}$$

with

- $a=\dfrac{0.45724R^2T_c^2}{p_c}$
- $b=\dfrac{0.07780RT_c}{p_c}$
- $\alpha=(1+\kappa(1-T_r^{0.5})^2$
- $\kappa=0.37464+1.54226\omega-0.26992\omega^2$ when ω < or = 0.49
- $\kappa=0.379642+1.48503\omega-0.164423\omega^2+0.016666\omega^3$ when ω>0.49
- $T_r=\dfrac{T}{T_c}$ noting that ω,κ,α are dimensionless

In more compact and dimensionless form, the **Peng-Robinson EOS** can be re-stated as a cubic in Z using

$$A=\frac{\alpha a p}{R^2T^2} \text{ and } B=\frac{bp}{RT}$$

$$Z^3-(1-B)Z^2+(A-2B-3B^2)Z-(AB-B^2-B^3)=0 \tag{5.5}$$

At a given temperature, T, and for fluid physical properties T_c, P_c, and ω (from **Part III**), and where P is the vapor pressure calculated for that fluid by the VP equation of **Part IV**, dimensionless parameters "A" and "B" are determined for Equation (5.5). This sets up a cubic equation in Z, which has three real roots (as opposed to being imaginary). The largest root is the value of Z_v; the smallest root is the value of Z_L; and the third root is discarded as having no physical meaning.

Then knowing Z_v and Z_L, the value of ΔZ can be calculated, from which the latent heat of vaporization (at that T & P) could be calculated using *Equation 4-1*. Also from Z_v and Z_L, the saturated phase densities can be calculated as estimates. But again, the P-R EOS will not hold up under this much mathematical manipulation to give any more than rough estimates of these distillation design properties; and so it is recommended to only use the P-R EOS to estimate fugacities (unless rough estimates of latent heat and phase densities are acceptable) . While the values for saturated vapor-phase densities appear to be quite reasonable up to 80% of critical pressure, the estimated values for saturated liquid-phase densities are generally found to be rather poor. To get an idea of this inaccuracy, compare the ambient temperature liquid density value to the EOS value for saturation pressure equivalent to ambient temperature. **For a good work-around to this problem, see the attachment "Saturated Liquid Densities.pdf". For a good work-around to the inaccuracy of duplicating the latent heat of vaporization at near-ambient pressures, see the attachment "Heat of Vaporization.pdf".**

A cautionary statement is in order regarding the use of an EOS (whether Van der Waals, SRK, P-R or others) with liquid mixtures. In some hydrocarbon applications, VLE convergence calculations attempt to combine the critical properties of a mixture's components, as a "psuedo-critical" mixture property, and use these "psuedo-criticals" in EOS calculations along with various mixing models. While there has been some success in this approach for certain hydrocarbons (mostly in the oil and gas industry), such approach rarely gives good results with inorganic polar or mildly polar mixtures, such as chlorosilanes. Various papers have been issued over the recent years promoting such approach, suggesting the use of "binary interaction parameters" to estimate an EOS for a mixture. There are simply too many differences in polarity, acentric factor, and individual fluid critical properties for this to work. Therefore, the reader is discouraged from such practice. Sometimes calculations must be worked out, checked along the way, and results validated - rather than taking an "easy way out". Certainly with computer automation, there is no reason not to do the full calculation procedure.

One stumbling block to many is that solving the EOS equation for Z means solving a cubic equation. Currently there are no MS Excel functions for solving the roots of a cubic equation. However a ready solution is found in an older CRC "Standard Math Tables" book, which can be solved with a spreadsheet or programmed into a macro.

Cubic Equations

Method for solving for the roots of a cubic equation in the variable "y", with roots y_1, y_2 and y_3

$$y^3 + py^2 + qy + r = 0 \tag{5.6}$$

may be transformed to the format , $x^3 + ax + b = 0$ by substituting for y the value, $x - \frac{p}{3}$. Where $a = \frac{1}{3}(3q - p^2)$ and $b = \frac{1}{27}(2p^3 - 9pq + 27r)$

If p, q, and r are real, then

- If $\frac{b^2}{4} + \frac{a^3}{27} > 0$, there will be one real root and two conjugate imaginary roots.
- If $\frac{b^2}{4} + \frac{a^3}{27} = 0$, there will be three real roots of which at least two are equal.
- If $\frac{b^2}{4} + \frac{a^3}{27} < 0$, there will be three real roots and unequal roots.

For saturated liquids and vapors, the last case is always true so a trigonometric solution is useful for solving EOS equations. Compute the value of the angle ϕ in the expression

$$\cos\phi = \frac{-b}{2} \div \sqrt{\left(\frac{-a^3}{27}\right)} \tag{5.7}$$

Then the three transformed roots x_1, x_2, and x_3 will have the following values:

- $x_1 = 2\sqrt{\frac{-a}{3}} \times cos(\frac{\phi}{3})$ and so cubic root $y_1 = -\frac{p}{3} + x_1$
- $x_2 = 2\sqrt{\frac{-a}{3}} \times cos(\frac{\phi}{3} + 120^o)$ and cubic root $y_2 = -\frac{p}{3} + x_2$
- $x_3 = 2\sqrt{\frac{-a}{3}} \times cos(\frac{\phi}{3} + 240^o)$ and cubic root $y_3 = -\frac{p}{3} + x_3$

There is an alternate algebraic solution to solving a cubic equation (which is frequently needed) to evaluate Z for superheated vapors. But for liquids and vapors near saturation, this alternate solution frequently breaks down close to 80-85% of critical pressure using MS Excel, due to software problems. Hence, the recommendation of the more stable trigonometric method, even though its math is seemingly odd. The trigonometric root-solving technique can break down somewhere near 95% of critical, using MS Excel because of numerical precision; but since the P-R EOS is only valid to 90% of critical, that is not a problem. Note that there are still other cubic root solving algorithms, but these also require even higher levels of numerical precision and are often found to break down (e.g., division by zero or requiring math with imaginary numbers).

When doing the above calculations, note that $\cos^{-1}$(ø) is only valid over the range between -1 and 1, returning values of ø between 1 to 0 radians, respectively. When doing automated calculations (via a spreadsheet like MS Excel), it is a good idea to check to see if the quantity $-b/(2\sqrt{-a^3/27})$ is between -1 and 1, in order to avoid an error message.

In Equation (5.5) above, the cubic equation's Z^3 term is unity, the Z^2 is $[-(1-B)] = B - 1$, the Z term is $[A\text{-}2B\text{-}3B^3]$ and the constant is $[-(AB - B^2 - B^3)]$. Then the solution of that cubic equation, using Equation (5.6):

- $p = B - 1$
- $q = A - 2B - 3B^2$
- $r = -(AB - B^2 - B^3)$
- $a = (3q - p^2)/3$
- $b = (2p^3 - 9pq + 27r)/27$

Intermediate parameters "p" and "r" will usually be negative, and parameter "q" will be positive; and the quantity $\frac{b^2}{4} + \frac{a^3}{27}$ will always be negative for liquids and vapors at or near saturation: so there will be three unequal roots. Note that for superheated vapors or vapors above critical pressure (aka gases), the quantity $\frac{b^2}{4} + \frac{a^3}{27}$ can be >0, in which case there is only one real root for Z, and an alternate algebraic solution method will be needed (i.e., the trigonometric method breaks down) . In such case, Z will be >1.

To use the trigonometric method to get Z_v and Z_L, first solve for the angle ø (in radians) of Equation (5.6):

$\phi = cos^{-1}(-b/(2\sqrt{-a^3/27})$ and so the three roots, y_1, y_2, and y_3 will be:

$y_1 = -p/3 + 2\sqrt{-a/3} \times cos(\phi/3)$, $y_2 = -p/3 + 2\sqrt{-a/3} \times cos(\phi/3 + 2\pi/3)$ and $y_3 = -p/3 + 2\sqrt{-a/3} \times cos(\phi/3 + 4\pi/3)$

The largest valued root is Z_v, and the smallest valued root is Z_L, with the intermediate valued root being discarded as meaningless.

Since the math in **Part V** of both the EOS and the cubic solution might seem daunting, an example is provided with calculations.

Example 5.1

Continuing the example of Part IV, of TCS vaporizing at 3.50 *atm*, and 347.05°K = 73.9°C, determine the values of Z_v , Z_L , and of $\Delta Z = Z_v - Z_L$. From the values of Z_v and Z_L determine the saturated vapor and liquid densities and then from the value of ΔZ determine the value of the latent heat of vaporization at 3.50 *atm*.

Solution

From **Part III**, Table 3-3, the pertinent property values for TCS are: MW = 135.452; T_c=479.15°K; P_c=41.15 *atm* ; and ω=0.2090. T_b , V_c and Z_c are not needed for this example.

For use in the P-R EOS, at the examples temperature of 347.05°K, T_r= 347.05/479.15 = 0.7243.

Using Equation (5.5) above for the P-R EOS and plugging in the values for T, P, T_r, T_c, P_c, and ω:

$a = 0.45724 \times 82.057^2 \times 479.15^2/41.15 = 1.7177E7$

$b = 0.07780 \times 82.057 \times 479.15/41.15 = 74.336$

$\kappa = 0.37464 + 1.54226 \times 0.02090 - 0.26992 \times 0.2090^2 = .68518$

$\alpha = (1 + 0.68518 \times (1 - 0.7243^{0.5}))^2 = 1.2145@ : 347.05°K$

$A = 1.2145 \times 1.7177E7 \times 3.50/(82.057^2 \times 347.05^2)$=0.09003 @: 347.05°K, 3.50 *atm*

$B = (74.336 \times 3.50)/(82.057 \times 437.05) = 9.136E - 3$ @: 347.05°K, 3.50 *atm*

Then the cubic equation to solve (using Equation (5.6)) is $Z^3 + pZ^2 + qZ + r = 0$, with terms:

- $p = B - 1$
- $q = A - 2B - 3B^2$
- $r = -(AB - B^2 - B^3)$
- $a = (3q - p^2)/3$
- $b = (2p^3 - 9pq + 27r)/27$
- $p = 0.0091361 - 1 = -0.99086$
- $q = 0.09003 - 2 \times 0.00913161 - 3 \times 0.009136^2 = 0.071516$
- $r = -(0.09003 \times 0.009136 - 0.009136^2 - 0.009136^3) = -0.00073819$

so for the trigonometric solution a and b are:

$a = (3 \times 0.071516 - (-0.99086)^2)/3 = -.25575$

$b = (2(-0.99086)^3 - 9 \times (-0.99086) \times 0.071516 + 27 \times (-0.00073819))/27 = -0.049179$

$b^2/4 + a^3/27$ = -1.49326E-5, so three real roots for Z are confirmed

$\phi = cos^{-1}(-b/(2\sqrt{-a^3/27})$= 0.15588 radians and so the three roots are:

- $Z_1 = -p/3 + 2\sqrt{-a/3} \times cos(\phi/3) = 0.91345$
- $Z_2 = -p/3 + 2\sqrt{-a/3} \times cos(\phi/3 + 2\pi/3) = 0.012439$
- $Z_3 = -p/3 + 2\sqrt{-a/3} \times cos(\phi/3 + 4\pi/3) = 0.064968$

Z_v is the largest root = 0.91345 and $Z_L$is the smallest root = 0.01244 , with the Z_3 root being discarded as meaningless. Z_v and Z_L will be used in **Part VI** to give estimates of fugacity coefficients, which partially account for departures from ideal mixture behavior.

$\Delta Z = Z_v - Z_L$= 0.91345-0.01244 = 0.90101 (note that this is less than the unity value of **Part II**, Equation 2-3 and 2-4 , which led to simplified VP equations).

If Equation (5.1) is re-written as $\frac{1}{V} = \frac{P}{ZRT}$ that gives the density in molar units,

and then multiply by molecular weight to get density in more familiar mass units.

Predicted vapor density @ 3.50 *atm*, $73.9°C = 3.50/(0.91345 \times 82.057 \times 347.05)$ = 1.3455E-4 g-mole/cc . In mass units = (135.45 x 1.3455E-4) = 1.822E-2 g/cc

Predicted liquid density @3.50 *atm*,$73.9°C = 3.50/(0.012439 \times 82.057 \times 347.05)$ = 9.8804E-3 g-mole/cc . In mass units = (135.45 x 9.8804E-3) = 1.338 g/cc

Looking at some TCS saturated liquid phase density data, it can be seen that the predicted saturated liquid density by the P-R EOS is high by about 8%, with a somewhat greater inaccuracy for sub-cooled liquid TCS. So unless there is better data available, the P-R EOS should not be thought of as better than ±10% on saturated phase density. Comparable TCS saturated vapor phase density data does not seem to exist, so no accuracy estimates can be made. See also the attached file "Saturated Liquid Densities", which can alleviate the EOS' inaccuracy on liquid densities.

In order to estimate the latent heat of vaporization at these conditions, it is necessary to go back to **Part IV**, Equation 4.5 and the example values: T_r= 0.7243; A=9.9796; B_1=0.38395; B_2= -0.11827; B_3=0.063708; c=2.5687; n=5.7950; and k=0.0446.

$$\psi = \frac{\Delta H_\nu}{\Delta ZRT} = A(1 - B_1 T_r + B_2 T_r^2 - B_3 T_r^3 + c(T_r^n - k))$$

and substituting in the values from above, but using a value for R=1.9859 to get ΔH_v in energy units of calories/gram-mole;

$$\begin{gathered}\Delta H_\nu/(0.90101 \times 1.9859 \times 347.05) \\ = 9.9796 \times (1 - 0.38395 \times 0.7243 - 0.11827 \times 0.7243^2 - 0.063708 \times 0.7243^3) \\ + 2.5687(0.72435.7950 - 0.0446)\end{gathered}$$

So ΔH_v is predicted at = 4,114 cal/g-mole. Multiplying by the MW, the latent heat of vaporization is predicted as 30.4 cal/gram at 3.50 *atm* and 72.9°C. Better estimates of the TCS latent heat at these conditions by previous work done under NASA contract indicated a value closer to 50 cal/gram (but these were still estimates, not direct measurements, and are possibly in error as well).

This illustrates the potential power of using an EOS with a thermodynamically consistent VP equation, but also shows the degree of inaccuracy of pairing the P-R EOS with the VP derivative of **Part IV**, Equation 4.5. In this case, it is unknown how much of the inaccuracy is caused by the EOS, and how much caused by VP equation’s derivative.

As will be seen in **Part VI**, the best use of the P-R EOS is just to estimate fugacity coefficients, which are relative quantities rather than absolute.

6: Fugacity

Distillation Science (a blend of Chemistry and Chemical Engineering)

This is **Part VI, Fugacity** of a ten-part series of technical articles on Distillation Science, as is currently practiced on an industrial level. See also **Part I, Overview** for introductory comments, the scope of the article series, and nomenclature.

Part VI, Fugacity deals with the expected departure of apparent pure-component vapor pressure in the vapor-liquid equilibria (VLE) of binary mixtures, as is commonly found in practical application of distillation science. **Part VII, Liquid Activity Coefficients** builds on this for mixtures that have greater-than-expected departures from ideal behavior.

This article uses the Vapor Pressure of **Part IV** and Equations of State of **Part V**.

The ideal vapor-liquid behavior of volatile fluid mixtures is governed by the combination of Raoult's Law and Dalton's Law. Neither of these is a scientific law but rather a representation of ideal behavior, closely approximated when the fluids in question have very similar properties. Raoult's Law of liquid partial pressures ($PP_i = X_i * VP_i$) states that the partial pressure of a fluid is the liquid mole fraction times the pure-component vapor pressure. Dalton's Law ($Y_i = PP_i / \Sigma\{PP_i\}$) states that the vapor mole fraction of a volatile component is that component's partial pressure divided by the sum of all partial pressures. If either of these two relationships is extended, with either X_i or Y_i being zero or unity, a tautology results. The nomenclature used here is: X_i and Y_i are the liquid and vapor mole fractions of the *ith* component in the mixture; VP_i and PP_i are the vapor pressure and partial pressure of the *ith* component.

Combining Raoult & Dalton, for a binary mixture:

$$Y_1 = \frac{X_1 \times VP_1}{X_1 \times VP_1 + X_2 \times VP_2} \tag{6.1}$$

$$Y_2 = \frac{X_2 \times VP_2}{X_1 \times VP_1 + X_2 \times VP_2}$$

However, in actual practice this never exactly works out. The more volatile fluid exerts an effect on the less volatile fluid, seemingly boosting that fluid's vapor pressure slightly. Conversely, the less volatile fluid slightly reduces the more volatile fluid's vapor pressure. There are also interactions in the vapor phase due to compressibility (Z_V, from **Part V**), which are sometimes significant. These departures from ideal behavior are due to two factors: (1) the fluid's phase change is at a temperature that is above or below its pure components' boiling points; (2) interactions occur between molecules with different properties. From a distillation column design perspective, these departures are important, since they make the component separations more difficult.

The technical term for the first factor of departure is fugacity, which has the same units as the fluid's vapor pressure, including the slight positive or negative secondary effect. The ratio between a liquid's fugacity and its vapor pressure is the liquid-phase fugacity coefficient; between the vapor's fugacity and its partial pressure is the vapor-phase fugacity coefficient. For a fluid by itself, fugacity has no meaning.

But in a mixture of two volatile fluids, it is the vapor and liquid fugacities that are in equilibrium with real fluids (as opposed to a mixture of ideal fluids, such as with Raoult/Dalton).

$$F_i^L = \phi_i^L \times VP_i \tag{6.2}$$

where $ø^L_i$ denotes the liquid fugacity coefficient

$$F_i^V = \phi_i^V \times PP_i \tag{6.3}$$

where $ø^V_i$ denotes the vapor fugacity coefficient

Consider a mixture of "fluid 1" and "fluid 2". At a given temperature, "fluid 1" has a pure-component vapor pressure of VP_1. But the presence of "fluid 2" alters that somewhat, so "fluid 1's" fugacity (F_1) is different from its pure-component vapor pressure. Conversely the presence of "fluid 1" somewhat alters "fluid 2's" fugacity (F_2), away from it's pure-component vapor pressure of VP_2. The higher the mole fraction of "fluid 2", the more "fluid 1's " fugacity is altered away from its pure-component vapor pressure. And vice-versa.

This is how fluid mixtures behave in the real world: Raoult/Dalton is just a introductory concept that would apply in an ideal world.

> The technical term for the second factor of departure is the **Liquid Activity Coefficient**, which is covered extensively in Part VII, but initially mentioned here briefly for a more complete understanding.

Fugacity effects are always present whenever two fluids of different volatility are mixed and vaporized or condensed. The Liquid Activity Coefficient effects are always present, but more noticed when the two fluids have significantly different properties, and even more so when the fluids are polar.

This is best illustrated by an example, partially continued from **Part IV** and **Part V**. At 73.90°C = 347.05°K, TCS has a pure component vapor pressure of 3.50 atmospheres. At that same temperature, STC has a pure component vapor pressure of 1.65 atmospheres. In this example, the liquid mixture is 40% mole fraction TCS, so X_1= 0.40 . Therefore STC's liquid mole fraction, X_2= 0.60. *Note: it is conventional notation for the most volatile fluid to have the lowest subscript.*

Per Raoult, the partial pressures PP_1 and PP_2 should be $0.40 \times 3.50 \ = 1.40$ and $0.60 \times 1.65 \ = 0.991$ atmospheres, respectively. This liquid mixture at 73.90°C would be expected to boil at $0.40 + 1.99 \ = 2.39$ atmospheres total system pressure; with the TCS mole fraction of the first bubble's vapor (Y_1) being $1.40/2.39 = 0.586$= 0.586 (so Y_2 is 0.414).

Instead, it is found that first bubble's vapor has a TCS mole fraction (Y_1) of only 0.573 (with Y_2 being 0.427) indicating that the TCS was just a little less volatile than expected and the STC was just a little more. While there is just over 2% difference in this case between Raoult/Dalton's expected and the actual resulting vapor mole fractions, the slight difference is truly present. If the difference in vapor pressures between the two fluids increased, and the overall pressure increased towards critical, the departure from the Raoult & Dalton relationship would then increase.

Two departure-from-ideality factors were purposefully shown in the above example: Fugacity and Liquid Activity Coefficients. The vapor-phase and liquid-phase fugacity coefficients (ϕ_i^V& ϕ_i^L) are never exactly unity (as implicit with Raoult &Dalton), but they do often approach unity (especially at low pressures and for molecularly similar fluids). Likewise, the Liquid Activity Coefficients (γ_i) are never exactly unity for real fluid mixtures (see **Part VII**), but are often set to unity because the fluids are non-interactive. The point to make here is that non-unity fugacity coefficients are just a result of fluids having different volatilities (i.e., they are not fluid-dependent); whereas Liquid Activity Coefficients are specific fluid-system dependent (e.g., TCS-STC has a different set of parameters than DCS-TCS). (*There is no analog to Liquid Activity Coefficients in the vapor phase - vapors just interact due to fugacity effects*.)

In the above example, the overall deviation of 2.3% (0.586 vs 0.573) on TCS mole fraction is almost all due to fugacity (both liquid and vapor), with a lesser amount due to Liquid Activity Coefficients. When designing a distillation for high-reflux purification, the relative importance between these two departure factors often reverses, as well as the overall magnitude.

The general equation (including the γ_i covered in **Part VII**), is that:

$$Y_i \times \pi = PP_i = (\phi_i^L/\phi_i^V) \times \gamma_i \times VP_i \times X_i \tag{6.4}$$

where π is the total pressure

Setting aside further discussion of Liquid Activity Coefficients until **Part VII**, the remainder of this article deals with how to determine the fugacity coefficients,ϕ_i^V & ϕ_i^L. That determination is dependent on the constants of **Part III**, and the calculation of the compressibility factors Z_v and Z_L from **Part V**.

For either phase, the equation for fugacity coefficient, as derived from thermodynamics in integral form is:

$$\mathrm{Ln}(\varnothing) = \int_0^P \left[\left(\frac{V}{RT}\right) - \frac{1}{P}\right] dP = \int_0^P \left[Z - \frac{1}{P}\right] dP \tag{6.5}$$

Fortunately, the Peng-Robinson Equation of State (P-R EOS) is good enough for most distillation design to evaluate fugacity coefficients based on some simplifications (see **Part V**). Admittedly the P-R EOS does not exactly fit chlorosilanes (and similar fluids), and a better EOS would be preferable, if it exists. However P-R is adequate to the task, especially with the "work-arounds" given below.

To evaluate ϕ_i^V there is an almost linear relationship between Z_v and P until close to P_r=0.4. So that allows Equation 6-5 to be reduced to

$$Ln(\phi) = (Z_V^{T,P} - Z_V^{sat}) \tag{6.6}$$

where $Z_V^{T,P}$ and Z_V^{sat} are respectively the component's vapor-phase compressibility at the temperature and pressure of the boiling mixture, and at the pure-component saturation condition (T & VP). For P_r>0.4, Equation 6-5 tends to give low values of ϕ^{ν}.

Since many industrial processes operate at higher pressures (closer to critical than P_r=0.4) another "work-around" is needed. Between 0.3< P_r<0.90, a series expansion cubic polynomial fits well for Z_v as a function of P_r, which then expresses the above thermodynamic integral form $(Z_v-1)/P_r$ to the function:

$$Ln(\phi^{\nu}) = a_3(P_{r2}^3 - P_{r1}^3) + a_2(P_{r2}^2 - P_{r1}^2) + a_1(P_{r2} - P_{r1}) + a_0[(LnP_{r2}) - Ln(P_{r1})] \tag{6.7}$$

where P_{r2} and P_{r1} are the reduced pressures of the mixture boiling pressure and the component's saturation pressure. The constants for Equation (6.7), a_3 through a_0, are -0.175248; 0.393866; -0.887714; and -0.0141409, respectively.

For ϕ_i^L, the evaluation of is simpler, since Z_L is fairly insensitive to pressure, simplifying the relationship for the *ith* component to be:

$$Ln(\phi^L) = [Z_L \times (\pi - VP)/VP] \tag{6.8}$$

where π is the system total pressure, VP is the pure component's vapor pressure and Z_L is the compressibility at the component's saturation temperature/pressure.

To the extent that the system pressure is greater than the *ith* component's vapor pressure (i.e., that liquid component is forcibly sub-cooled), the value of ϕ_i^L will be greater than unity. To the extent that the system pressure is less than the *ith* component's vapor pressure (i.e., that liquid component is forcibly super-heated), the value of ϕ_i^L will be less than unity.

The pattern for ϕ_i^V is usually the reverse as a result of compressibility changes, so the value of ϕ_i^L/ϕ_i^V in Equation (6.4) above shows how the distinct difference in Raoult/Dalton departure results. In the example above (40% molar TCS in STC @ 73.9°C saturation), the values are given in Table 6-1:

Table 6-1: Comparing TCS and STC Fugacity Coefficients

	TCS	STC
ϕ^L	0.9976	1.0039
ϕ^V	1.0488	0.9562
ϕ^L/ϕ^V	0.9512	1.0498

The careful reader will note an apparent discrepancy between the overall 2.3% ideality departures reported in the example above and the fugacity ratio of Table 6-1. Just the fugacity effects should make the TCS mole fraction departure about 4.8% lower than Raoult/Dalton expectation, but the Liquid Activity Coefficient effects restores about half of the departure. Thus it can be seen that fugacity effects and Liquid Activity Coefficient effects can be counteractive.

The "take-away" from this article is:

- Raoult & Dalton describe ideal behavior of vaporizing liquid mixtures, but do not completely describe the behavior of real fluids
- All departure factors must be considered: liquid fugacity, vapor fugacity and Liquid Activity Coefficients

7: Liquid Activity Coefficients

Distillation Science (a blend of Chemistry and Chemical Engineering)

This is **Part VII, Liquid Activity Coefficients** of a ten-part series of technical articles on Distillation Science, as is currently practiced on an industrial level. See also **Part I, Overview** for introductory comments, the scope of the article series, and nomenclature.

Part VII, Liquid Activity Coefficients builds on **Part VI, Fugacity** regarding the departure of vapor-liquid equilibria (VLE) of binary mixtures from ideal behavior, as is commonly found in practical application of distillation science. In conjunction with previous articles, the goal of this article to to complete the explanation of equilibrium behavior of binary systems; such that **Part IX** can illustrate an example of a distillation process design.

In **Part VI** the combination of Raoult's Law and Dalton's Law was introduced governing the ideal vapor-liquid behavior of volatile fluid mixtures, with ***Equation 6-1*** repeated for convenience.

$Y_1 = (X_1 \times VP_1)/(X_1 \times VP_1 + X_2 \times VP_2)$ $Y_2 = (X_2 \times VP_2)/(X_1 \times VP_1 + X_2 \times VP_2)$ **repeat Equation 6-1**

Also in **Part VI** the real-world departure from ideal behavior was discussed as far as the fugacity coefficients, but stopped short of Liquid Activity Coefficients. *Equation 6-4* repeated below gave the complete relationship between the partial pressure of the *ith* component, PP_i, and the pure-component vapor pressure VP_i , for liquid and vapor mole fractions X_i and Y_i, respectively. The total system pressure (sum of the partial pressures) is denoted as π; (ϕ_i^V& ϕ_i^L) are the liquid and vapor fugacity coefficients. γ_i is the *ith* component's Liquid Activity Coefficient for that specific set of components.

$Y_i \times \pi = PP_i = (\phi_i^L/\phi_i^V) \times \gamma_i \times VP_i \times X_i$ **repeat Equation 6-4**

An example was given in **Part VI** of a vaporizing TCS-STC mixture, contrasting the vapor-liquid equilibria (VLE) expected from Raoult/Dalton; from what actually occurs in the real world as a result of the ideality departures. Summarizing the previous article, the more volatile fluid exerts an effect on the less volatile fluid, to increase its apparent vapor pressure; and the less volatile fluid reduces the more volatile fluid's vapor pressure. The net effect is to make the component separations more difficult via distillation (e.g., more theoretical trays or increased amount of reflux than expected). For detailed discussion of fugacity coefficients, see **Part VI**.

As a "house-keeping" note, it must be emphasized that **Parts VI** and **VII** treat fugacity coefficients and Liquid Activity Coefficients separately. In some texts, the two topics are compressed together using specific models of mixing and Equations of State (EOS). However, the two ideality departure factors are differently based, and their combination often gives erroneous or nonsense results when used with polar fluids, or those that have a high degree of hydrogen bonding (such as the chlorosilanes that are used as continuing examples). Fugacity coefficients naturally result from differences in fluid volatility. Liquid Activity Coefficients result from differences in fluid properties, including critical temperature (T_c), critical pressure (P_c), critical volume (V_c) and acentric factor (ω), as well as the entropy & enthalpy effects of mixing (referred to in thermodynamic texts as excess Gibbs Free Energy).

For a given temperature and combination of fluids, the value of γ is a function of the liquid mole fraction, rising asymptotically from unity at the pure component condition. Typically the values of γ for a system are plotted as a function of the more volatile component's mole fraction. An example of such a plot is given in Figure 7-1, for the TCS-STC binary at 73.9°C (the temperature of the continuing example of previous articles).

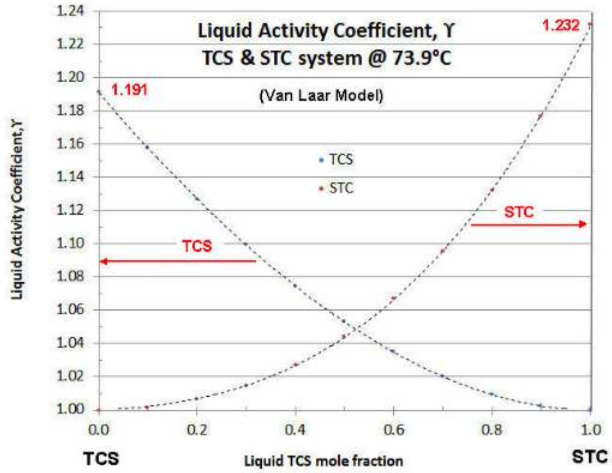


Figure 7-1, **TCS-STC Liquid Activity Coefficients @ 73.9°C**

There are several models for Liquid Activity Coefficients that are thermodynamically consistent (i.e., follow the Gibbs-Duhem Rule of binary system thermodynamics). The two-constant Van Laar model is the easiest to manipulate, but is limited to binary systems. The Margules model is not that different from the Van Laar model, and also limited to binary systems. The Wilson model is more complex mathematically, but can be applied to ternary or greater-numbered component systems. Fortunately the Wilson model parameters can be calculated from the Van Laar model. Even more complex Liquid Activity Coefficient models are known (e.g., NRTL and UNIQUAC), but these require more data points to effectively evaluate. In the case of the above plot, the Van Laar model is shown. In **Part VIII**, the techniques are discussed to evaluate experimental data on Liquid Activity Coefficients and fit the data to a model. In most cases with chlorosilanes (and their impurities) the data quality is not that great, so the simpler Van Laar is typically used to correlate data and for evaluating binary systems, and the Wilson model is used for ternary systems and beyond. For a more in-depth discussion of Liquid Activity Coefficients, the reader is referred to the text " The Properties of Gases and Liquids", by Reid, Prausnitz and Sherwood, (McGraw-Hill). The third edition is more readable, but the fifth edition is more updated.

The Van Laar model is:

$$Ln(\gamma_1) = A_1 \times [1 + (A_1 \times X_1)/(A_2 \times X_2)]^{-2} \tag{7.1}$$

$$Ln(\gamma_2) = A_2 \times [1 + (A_2 \times X_2)/(A_1 \times X_1)]^{-2}$$

with A_1 and A_2 being the Van Laar constants for the more volatile and less volatile component, respectively.

Conveniently,

$$EXP(A_1) = \gamma_1 @ X_2 \rightarrow 0 \tag{7.2}$$

$$EXP(A_2) = \gamma_2 @ X_1 \rightarrow 0$$

which simplifies evaluation. Note from Figure 7-1 that the Liquid Activity Coefficients plot is not always symmetrical (i.e., $A_1 \neq A_2$). In the plot of Figure 7-1, A_1 (TCS) = 0.1752, so the TCS curve asymptotes at γ= 1.191; and A_2 (STC) = 0.2086, so the TCS curve asymptotes at γ= 1.232 .

Completing the on-going example from **Part VI**, for a TCS-STC liquid mixture, with 40% molar TCS, at 73.9°C, to determine the vapor mole fraction in equilibrium (but now including the Liquid Activity Coefficient effects):

In **Part VI**, the pure-component vapor pressures of TCS and STC respectively were calculated from the VP equation of **Part IV** to be 3.500 and 1.651 atmospheres, respectively at 73.9°C. Also from **Part V**, the liquid and vapor fugacities were calculated as per Table 6-1, repeated below, using the Peng-Robinson EOS (**Part V**).

Repeat Table 6-1: Comparing TCS and STC Fugacity Coefficients

	TCS	STC
ϕ^L	0.9976	1.0039

	TCS	STC
ϕ^V	1.0488	0.9562
ϕ^L/ϕ^V	0.9512	1.0498

The values for γ are calculated for TCS and STC @ 0.40 mf TCS, per Equation (7.1) using the values for A_1 and A_2 above (or just read from Figure 7-1), to be 1.075 and 1.027 respectively. Plugging all these values into Equation 6-4 (repeated above), the partial pressures of TCS and STC @ 73.9°C are:

$PP_{TCS} = 0.9512 \times 1.075 \times 3.500 \times 0.40 = 1.432$ atmospheres, and

$PP_{STC} = 1.0498 \times 1.027 \times 1.651 \times 0.60 = 1.068$ atmospheres.

So the value of $Y_{TCS} = 1.432/(1.432 + 1.068) = .573$, which was given in Part VI, with the STC mole fraction being 0.427.

The careful reader will see that not only are the partial pressures of TCS and STC different from Raoult/Dalton, but so is the total pressure. In the ideal behavior of Raoult/Dalton, the partial pressures were given in **Part VI** as 1.400 and 0.991 atmospheres, for a total pressure of 2.39 atmospheres. But in reality, at 73.9°C and 0.40 mf TCS, the total pressure would instead be 2.50 atmospheres. So while the ideality departures of TCS made the vapor mole fraction smaller (0.573 vs 0.586), both the TCS and STC exerted slightly more partial pressure than Raoult would indicate. But the TCS mole fraction still went down, because the ideality departure of STC was greater than that of TCS.

The take-away from this example is that while Raoult/Dalton will give you a "quick & dirty" value for Y/X, it is easy to get the overall wrong answer for determining the requirements of a distillation column design. In almost every instance of industrial practice, the use of Raoult/Dalton will undersize the number of trays needed to make a separation (or for the same number of trays, undersize the reflux required). For this reason, it is important to know how to work out the better answer.

For many binary systems, VLE data exists – but typically not at the temperature/pressure needed for industrial design. Most industrial designs use higher pressures for economy of size, as well as to accommodate available energy sources for driving the column reboiler and condenser. Yet the practicalities of data collection in the laboratory usually require pressures slightly above ambient, or at slight vacuums. This is especially true of chlorosilanes, since the normal materials of construction for higher pressures can catalyze slow side reactions, such as disproportionation and dimerization.

Lab data is normally collected based on small amounts of fluid that are used repeatedly at somewhat different combinations of temperature and composition. In the case of electronic impurities, some of these compounds are not very stable outside of a chlorosilane matrix. So a correlating method must be used that allows a certain degree of extrapolation.

In 1981 Chung-Ton Lin and Thomas Daubert from Penn State University developed such a Liquid Activity Coefficient model and published same in Industrial & Engineering Chemistry Process Design and Development, basing their work on non-polar hydrocarbon mixtures. I have used their general method, but modified two of the constants so as to better fit available VLE data on chlorosilanes. Using these re-evaluated constants, the revised "Lin & Daubert" method (detailed below as Equation (7.3) was used to check against industrially obtained data on the distillation purification of TCS and silane, with good results. Obviously more data would be preferable, especially on ternary mixtures and low-concentration electronic impurities in chlorosilanes. However until such becomes available, Equation (7.3) below is recommended.

To develop a generic expression for the Van Laar parameters (A_i & A_j), Lin & Daubert go back to Van Laar's original two-term expression regarding the excess Gibbs energy of mixing, using the SRK Equation of State for partial molar compressibility and simple mixing rules; and expand it into the following:

$$A_i = c \times F_i + d \times k_{ij} \times G_i \tag{7.3}$$

$$A_j = c \times F_j + d \times k_{ij} \times G_j$$

- $F_i = 1/[T_{ri}P_{ci}] \times (R_i \times (1+M_j)^{0.5} \times P_{ci}^{0.5} - R_j \times (1+M_j)^{0.5} \times P_{cj}^{0.5})^2$
- $F_j = 1/[T_{rj}P_{cj}] \times (R_j \times (1+M_i)^{0.5} \times P_{cj}^{0.5} - R_i \times (1+M_i)^{0.5} \times P_{ci}^{0.5})^2$
- $G_i = T_{ri}^{-1}(P_{cj}/P_{ci})^{0.5} \times R_i \times (1+M_i)^{0.5} \times R_j(1+M_i)^{0.5}$
- $G_j = T_{rj}^{-1}(P_{ci}/P_{cj})^{0.5} \times R_j \times (1+M_j)^{0.5} \times R_i(1+M_j)^{0.5}$
- $M_i = 0.480 + 1.574\omega_i - .176\omega_i^2$ $M_j = 0.480 + 1.574\omega_j - .176\omega_j^2$
- $R_i = [1 + M_i \times (1 - T_{ri}^{0/5})]^{0.5}$ $R_j = [1 + M_j \times (1 - T_{rj}^{0/5})]^{0.5}$

- $k_{ij} = 1 - 2 \times [V_{ci}^{1/3} \times V_{cj}^{1/3}]^{0.5} / [V_{ci}^{1/3} + V_{cj}^{1/3}]$

Using an improved set of curve-fits for "c",

- $0 < \Delta\omega < 0.03 \Rightarrow Ln(c) = -8.0637E + 04(\Delta\omega)^3 + 8.2649E + 03(\Delta\omega)^2 - 3.2891E + 02(\Delta\omega) + 5.5051$
- $0.03 < \Delta\omega < 0.30 \Rightarrow Ln(c) = 6.4645E + 02(\Delta\omega)^4 - 7.9857E + 02(\Delta\omega)^3 + 3.4768E + 02(\Delta\omega)^2 - 6.5099E + 01(\Delta\omega) + 2.5667$

Reducing available VLE data to Van Laar parameters, and adjusting for vapor and liquid fugacity coefficients, the best value of "d" is found to be 112.1 for chlorosilanes and similar fluids.

Lin & Daubert suggested that the "c" function be capped at $0.005 < \Delta\omega = |\omega_i - \omega_j| < 0.15$, based on their data; but using the above improved set of polynomials for "c" avoids singularities during iterative calculations, with a minimum "c" of 0.135 (as "Δω" → ∞)

Note that the "c" variable in the first term of Equation 7-3 is based on the F_i values and the differences between the two fluid's acentric factor, ω (i.e., their molecular complexity); whereas the second term of Equation 7-3 is based on the G_i values and the differences between the two fluid's molecular volumes at critical (V_c).

Returning to the continuing previous TCS-STC example, and using the two components' critical temperature (T_c), critical pressure (P_c), critical volume (V_c) and acentric factor (ω), as well as basing the reduced temperature (T_r) on 73.9°C = 347.05°K, Equation (7.3) gives the values of A_1 and A_2 used earlier in this article, of 0.1752 and 0.2086 respectively.

Frequently in evaluating different distillation designs for a process, the designer is faced with the need to make process temperature changes and determine the effect on Y/X using Equation 6-4, repeated above. That causes no problem in the evaluation of the fugacity ratio,ϕ_i^L/ϕ_i^V, since those factors are derived from the EOS (see **Part VI**). However, if using experimentally obtained values for γ that were obtained at a different temperature, there needs to be further adjustment. The general "rule-of-thumb" seen in some texts is that Ln(γ) is proportional to 1/T (absolute temperature) for small temperature changes. In reality such a "rule-of-thumb" is tantamount to making an assumption that the mixing effects of the multi-component fluid are closer to an isothermal model than an isenthalpic one. In most cases, the temperature dependency on activity coefficient is a blend of two theoretical models, which leads to the relationship:

$$Ln(\gamma_i) = a + \frac{b}{T} \tag{7.4}$$

where "a" and "b" are the constants of a linear relationship ("a" = 0 leading to the isenthalpic model, or "b" = 0 leading to the isothermal model). The use of Equation (7.3) solves the problem, since it predicts such temperature changes along the lines of Equation (7.4) , and so can be used in a relative manner. For example, if Equation (7.3) is found to over-predict the value of $Ln(\gamma)$ by 10% of an experimentally obtained activity coefficient value, then calculate how much Equation (7.3) would predict for the different temperature, and apply that 10% on the $Ln(\gamma)$.

Note that to all intents, the Liquid Activity Coefficient is not a direct function of pressure, other than increasing the boiling temperature increases pressure. However, fugacities are a function of both temperature and pressure, and that is why they should be included in Y/X calculations (many texts advocate setting fugacities to unity, which is only close to being valid at very low pressures).

To demonstrate how Equation (7.3) performs with changing temperature (from 330°K to 380°K), Figure 7-2 shows the dependency for the asymptotic BIP's of the TCS-STC mixture in the example above. Note that the slope of the STC Liquid Activity Coefficient line is about 20% steeper than that of the TCS Liquid Activity Coefficient line (i.e., in the TCS-STC binary system, increasing the temperature has a greater effect on the STC's γ than it does on the TCS's γ).

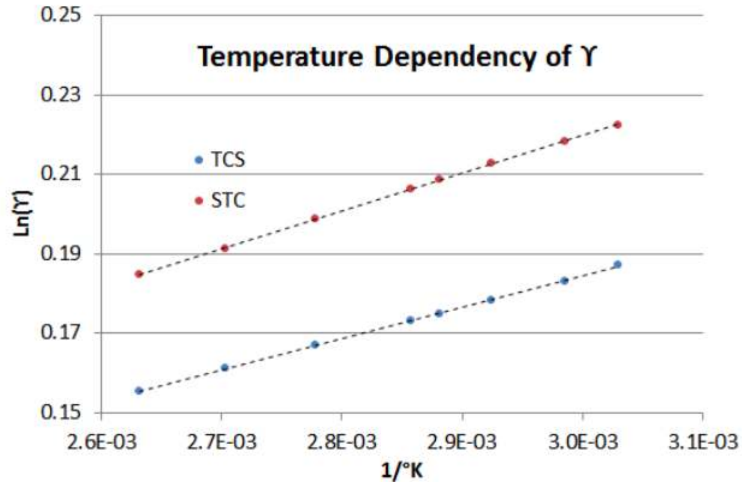


Figure 7-2, Temperature dependency of Ln(ϒ) at the asymptotes

8: VLE Analysis Methods

Distillation Science (a blend of Chemistry and Chemical Engineering)

This is **Part VIII, VLE Analysis Methods** of a ten-part series of technical articles on Distillation Science, as is currently practiced on an industrial level. See also **Part I, Overview** for introductory comments, the scope of the article series, and nomenclature.

Part VIII, VLE Analysis Methods recommends methodology used when data-collecting the Liquid Activity Coefficients of binary systems (see **Part VII)**. This article also deals with validation, especially when data collection is done on reactive fluids like chlorosilanes which can disproportionate or dimerize during study.

This article uses the information of **Parts III through VII**.

In **Part VII** a correlation technique was given for estimating Liquid Activity Coefficients. As mentioned in the previous article it is common to collect such vapor-liquid equilibria (VLE) data in the lab, at conditions close to ambient pressure. Yet many industrial applications frequently need to use VLE results at higher process pressure/temperatures or different compositions. To avoid notation confusion between the liquid activity coefficient (γ_i) and the vapor mole fraction ($\mathbf{Y}_i$), the mole fractions **X** and **Y** are shown as bolded.

Recall from **Parts VI and VII** that fugacity coefficients (ϕ_i^V& ϕ_i^L) are functions of temperature and pressure. However, Liquid Activity Coefficient s (γ_i) are functions of only temperature and mole fraction. In an experimental situation, the calculated values of γ_i (from the values of $\mathbf{Y}_i$ and $\mathbf{X}_i$ = vapor-phase mole fraction and liquid-phase mole fraction) are easiest to correlate when system temperature is held constant, and system pressure is allowed to vary. Stated otherwise, P**XY** data @ constant T is far easier to collect and correlate than T**XY** @ constant P. However, in commercial applications the distillation column is typically controlled at constant pressure. So a common industrial communication error is to request the lab to collect VLE data as T**XY**@P, but instead get P**XY**@T instead. There is a way to convert one data set to the other relationship, but it is somewhat awkward and requires choosing activity coefficient model.

In Figure 7-1 of **Part VII**, the expected profile of γ vs **X** @ constant T is shown, and in Figure 7-2 of **Part VII** the expected profile of γ vs T @ constant **X** is shown, at the **X**=0 and **X**=1 axes. The reason for generating Liquid Activity Coefficients experimentally is normally to use the lab results for distillation column design, with the confidence that γ can be accurately known at any value of **X** or T, so as to establish the **Y**/**X** relationship up and down the column design. Using the principles of **Parts VI** and **VII**, these problems can be resolved.

Having established the experimental protocol, it is necessary to consider the impact of data collection equipment on the fluids being analyzed and to validate the quality of the sample fluids. For reactive fluids such as chlorosilanes, that means using pressure-capable glassware is the best choice (or nickel-lined steel as a second choice), since some of the alloying elements of stainless steel will slowly catalyze disproportionation reactions, which alter the sample compositions.

Also, it is important to validate the purity of the fluid samples, as opposed to blindly accepting the analysis of any accompanying supplier information. Again, with chlorosilanes (as typical of reactive fluids), the shipping container can catalyze side reactions (i.e., the supplier's COA was perhaps accurate when the sample was loaded into the container, but purity degraded during shipment). It is common for 99.99% pure TCS samples (under argon inerting) from reputable suppliers to end up having several percent DCS and several percent STC, along with a few tenths percent hexachlorodisilane and a few tenths percent hydrogen gas, when used just a few weeks later. A good practice before loading sample fluids into the lab equipment is to double-distill samples (discarding the "lights" and "heavies" fractions, and assuring that only the "heart's cut" fraction is used: whose boiling pressure is identical to the expected pure component vapor pressure).

With the above precautions, data is collected at 15-20 (or more) **X**,**Y** points, with some duplicates later in the run to establish experimental analysis accuracy and confirm the absence of systemic error (such as contamination). At least two pairs of data points should be collected close to X= zero and X= unity. To help confirm temperature dependency of activity coefficients, at least three sets of data (each at a different constant temperature or pressure) should be collected.

After data collection, the lab work is suspended, but then "number-crunching" is needed to confirm the data's validity before reporting or using the results. If data validity is not confirmed, the error must be found and lab work repeated. First, plot out the data set and check that both the P-**X** curve and P-**Y** curve , or T-**X** curve and T-**Y** curve show an identical intersection, at both zero and unity mole fraction axes; and that the axis intersections are exactly representative of each pure-component's known vapor pressure. If the **X** and **Y** curves do not intersect at the correct value on each axis, there is some systematic error in the data set that must be resolved before any more evaluation work is done. A corrupted dataset is likely to be of only minimal value in establishing activity coefficients; and is more often a reason to toss it all out and re-do the work, after the systemic error is resolved.

Using an algebraic modification of previous Equation 6-4, activity coefficients, the γ_i i are calculated for each P-**X** and P-**Y** data point.

$$\gamma_i = (\phi_i^V/\phi_i^L) \times (Y_i \times \pi)/(VP_i \times X_i) \tag{8.1}$$

where π is the total pressure, VP_i is the vapor pressure of the *ith* component, and (ϕ_i^V/ϕ_i^L) is the vapor/liquid fugacity ratio of that component.

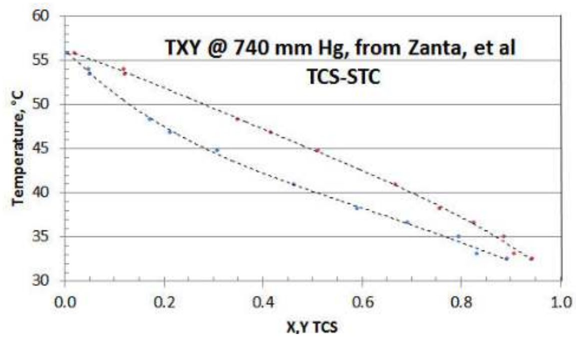


Figure 8-1, T**XY**/P VLE data on TCS-STC system

In Figure 8-1 the twelve-point T**XY**/P data set of Zanta et al, from Chemicky Prumysl (Czech Chemical Industry) is shown, using a spreadsheet plot. The researchers did collect a few data points near the STC axis (i.e., TCS= zero). However it would have been preferable to have a few more collected close to the other (i.e., TCS= unity) axis; and to have that all data reported with more precision. The researchers chose to collect data at a constant pressure of 740 mm Hg (as opposed to the more preferable P**XY**/T method that avoids the need for adjustment of calculated Liquid Activity Coefficient γ for changing temperature), for reasons of their lab equipment and simplifying their procedure; and to only collect one T**XY**/P data set.

If only the raw data is considered, it would appear that the T-**X** and T-**Y** curves might not mutually intersect at the STC axis, or that the intersection is higher than the expected 56.2°C temperature representative of STC vaporization @ 740 mm Hg. The reason is that individual data point precision is low in the region of the STC axis.

However, with data-smoothing and extrapolation of the T-**X** and T-**Y** curves via some spreadsheet curve-fitting, the two curves appear to quite likely both intersect the STC axis near the expected 56.2°C temperature. At the TCS axis however, curve-fitting the T-**X** and T-**Y** data appears to both result in their intersection on the TCS axis at 30.0°C, as opposed to the expected 32.3°C for pure TCS. The most likely explanation for this data set discrepancy is that the TCS sample supplied was either a disproportionated mix of DCS/TCS/STC as-received, and/or the fluids in the equipment disproportionated as a result of the equipment materials of construction. A TCS axis temperature of 30.0°C is representative of a 5% DCS/95% TCS liquid mixture. Surprisingly, the researchers' analytical procedure did not pick up the DCS peak, which would have revealed the sample corruption (but that is a common error using that the researcher's type of analyzer).

Now that the data set has been discredited, it remains to be seen what - if any- useful information can be extracted. Other than demonstrating technique, there is little use in reducing the data that is high in "bad TCS" to get the γ asymptote intersecting the TCS axis. That is because the $\mathbf{X}_{TCS*}$ mole fractions are changing along the STC activity coefficient curve (about a high 1:19 ratio of DCS/TCS, dropping in ratio to very little DCS in the ternary mixture, close to 56.2°C at the STC axis). It would be impossible to guess what property values are best to use for fugacity coefficients, given a "wild-card" DCS content). The data closest to intersecting the STC axis could be interpolated to make up a few "pseudo-points" close to $\mathbf{X}_{TCS}$=0, in order to get a rough value of the TCS γ asymptote (based on the assumption that there is little DCS in the ternary mix at that point). However, even small amounts of DCS near the $\mathbf{X}_{TCS}$=0 area could be expected to exert a significant Liquid Activity Coefficient effect, and the fugacity values might be somewhat in error by assuming only the property values of TCS.

However, to illustrate the data reduction technique for T**XY**/P data (as opposed to the preferred P**XY**/T form), a few "pseudo-points" are evaluated near the axis.

Using a fourth order curve-fit, with the STC axis temperature set to 56.20°C, the T-**X** and T-Y curve-fit equations are, respectively, via spreadsheet curve-fitting functions:

T-**X** $T = 18.949x^4 - 64.825x^3 + 75.709x^2 - 56.044x + 56.20$

T-**Y** $T = -22.243y^4 + 33.158y^3 - 18.028y^2 - 19.075y + 56.20$

From those two curve-fits, the following "pseudo-points" are calculated, by solving the quartic equations for T-**X** and T-**Y**:

Table 8-1, three "pseudo-points" made from curve-fits near the STC axis

P, mm Hg	T, °C	$\mathbf{X}_{TCS*}$	γ_{TCS*}
740	56.2	0.0000	0.0000
740	55.0	0.00359	0.0104
740	54.0	0.0415	0.1065
740	53.0	0.0620	0.1515

In Table 8-1, $\mathbf{X}_{TCS*}$ and $\mathbf{Y}_{TCS*}$ are subscripted as "TCS*" to denote that while TCS properties are used for vapor pressure and fugacity estimation, the more volatile fluid is really a DCS/TCS mix. The data-point at **X**, **Y** = 0.0000, 0.0000 is given just to show the STC axis intersection, but cannot be used to calculate the γ asymptote since that would result in "division by zero" in Equation 7-1.

Using the properties of TCS, the values for VP, $ø_L$ and $ø_v$ are calculated as per **Parts IV, V, and VI**, and the results shown in Table 8-2 for calculated γ_{TCS*} at 740mm Hg (for each "pseudo-point" temperature).

Table 8-2 for calculated γ TCS* at "pseudo-point" temperatures, from PXY/T data

P, mm Hg	T, °C	VP_{TCS*}	$ø_L$	$ø_v$	γ_{TCS*}
740	56.2	2.130	0.9983	1.0321	to be determined
740	55.0	2.055	0.9984	1.0316	1.418
740	54.0	1.994	0.9984	1.0311	1.294
740	53.0	1.934	0.9984	1.0307	1.270

The calculated values for γ_{TCS*} in Table 8-2 are based on the **X**,Y "pseudo-points" of Table 8-1 and using Equation (8.1) above, to demonstrate the methodology of experimental data reduction. The trending on fugacity coefficients and activity coefficient is correct: $ø_L$ values increase toward unity as $\mathbf{X}_{TCS*}$ increases toward unity; $ø_v$ values decrease toward unity as $\mathbf{X}_{TCS*}$ increases toward unity; γ_{TCS*} values decrease toward unity as $\mathbf{X}_{TCS*}$ increases toward unity.

The asymptote at the STC axis is determined by extrapolating value of γ_{TCS*} from the three "pseudo-points". (That curve would have a different slope with respect to composition if it were T**XY**/P data, since temperature is changing as well as $\mathbf{X}_{TCS*}$ in Table 7-2.) This technique identifies the γ_{TCS*} asymptote as 1.437 (the average value of γ_{TCS*} obtained by extrapolating curve-fits of γ_{TCS*} vs $\mathbf{X}_{TCS*}$ to a zero value of $\mathbf{X}_{TCS*}$, and γ_{TCS*} vs $\mathbf{Y}_{TCS*}$ to a zero value of $\mathbf{X}_{TCS*}$). Compared to other researcher's data, the calculated asymptote has a high value. It also seems high per expectation from Equation 7-3 from **Part VII**, which would indicate a value closer to 1.21 @ 56.2°C, or 1.23 @ 32.3°C (the expected TCS boiling point at 740 mm Hg). Possibly there is error from the DCS content, which would tend to make the calculated γ_{TCS*} values high by increasing the non-STC mole fraction in the vapor. The curve-fitting technique could be "off" since there was low precision in the two data pairs close to the X_{TCS*} =0 axis.

This demonstrates that there is simply no good way to "fix" data that has systematic error in it: virtually all texts suggest that when corrupted data is encountered, it is pointless to continue with data analysis. In addition to determining reasonable asymptotic values at either axis, Reid, Prausnitz, and Sherwood suggest in "*The Properties of Gases and Liquids*", that all γ calculated values be first adjusted to a common temperature basis (using the approximation of $Ln(\gamma) \times = constant$), then use the Gibbs-Duhem Law to establish thermodynamic consistency. While technically the best way to validate data for consistency, it requires a significant amount of high-quality data, and is suggested only if advanced Liquid Activity Coefficient models are to be considered (e.g., NRTL or UNIQUAC). For Van Laar, Margules, Wilson or similar two-constant models, 15-20 data points should be sufficient.

It is always better to have fewer data points, but have the dataset internally consistent, rather than a large number of possibly corrupted data points.

This article suggests using the Van Laar model for activity coefficients based on reasonable results with data analysis on chlorosilanes, and the success in predicting the Van Laar constants via Equation 7-3 in **Part VII**. However, the reader may want to explore other models for a better fit to experimental data, after such data has been validated. In “The Properties of Gases and Liquids”, the authors give an exhaustive list of other activity coefficient models to consider, with some notes on “pro & con”.

9: Putting It All Together

Distillation Science (a blend of Chemistry and Chemical Engineering)

This is **Part IX, Putting It All Together** of a ten-part series of technical articles on Distillation Science, as is currently practiced on an industrial level. See also **Part I, Overview** for introductory comments, the scope of the article series, and nomenclature.

Part IX, Putting It All Together uses the information from **Parts III through VII**, showing how to combine them in a practical example for bulk separation; and how using the techniques detailed in previous articles give answers that differ from the ideals of basic Raoult/Dalton Law application.

See previous articles per the following detail:

- **Part III** for critical properties and acentric factor, especially Table 3-3
- **Part IV** for the recommended vapor pressure equation, especially Equation 4-3 and Table 4-1
- **Part V** for Equation of State, especially Equations 5-3 **and** 5-4
- **Part VI,** especially Equation 6-4 for partial pressure and Equations 6-7 and 6-8 for fugacity coefficients
- **Part VII** for Liquid Activity Coefficients, especially Equation 7-3

In this article a practical distillation example is developed, showing the Y/X relationship with temperature for the DCS-TCS binary system with pressure is held constant at 11.0 atmospheres, and showing the results as a T**XY**/P plot as well as in tabular form. From such data, a distillation column designer would consider the best arrangement for column feed, “tops” and “bottoms” recovery, tray count and external reflux ratio (and from that, the energy required for the reboiler and condenser, as well as hydraulic loading per unit mass flow of feedstock). Two variations of this example are worked-up, showing the application of distillation science: one which only uses ideal Raoult/Dalton, and one that uses the full set of ideality departure coefficients. Table output and plots below are done using MS Excel. Then at the end of the article, the table results are graphically demonstrated using a McCabe-Theile plot to step off the trays for an appropriate reflux rate and show the best feed tray.

The table of Figure 9-1 is set up to cover the range of DCS liquid mole fraction from unity to zero, in increments of 0.05 mole fraction, with 0.01/0.99 mole fraction added to illustrate asymptotic effects. For each non-zero table row, a temperature is first assumed, and then the vapor mole fractions calculated per Raoult/Dalton, with p=11.0 atmospheres; then that row’s temperature is iteratively adjusted until the total pressure = 11.0 atmospheres.

In the plot of Figure 9-1 shows the calculation results in T**XY**/P form, with the value of $\mathbf{X}_{DCS}$=1.0 representing the tightest possible “tops” product at the distillation column condenser; and a value of $\mathbf{X}_{DCS}$=0.0 representing the tightest possible “bottoms” product at column reboiler, all at a total reflux condition. For a finite column, the mole fraction of the “tops” product would be a cropping of the upper $\mathbf{X}_{DCS}$ values of the T**XY**/P curves; and the mole fraction of the “bottoms” product a would be a cropping of the lesser $\mathbf{X}_{DCS}$ values of the T**XY**/P curves. In such instance, the column feed tray would be identified as that point in the column where the tray temperature is identical to the feed mixture’s boiling point. The accompanying plot of solution shows what the T**XY**/P curves would look like, for such a Raoult/Dalton solution to the DCS-TCS binary at 11.0 atmospheres.

$Y_1 = (X_1 \times VP_1)/(X_1 \times VP_1 + X_2 \times VP_2)$ $Y_2 = (X_2 \times VP_2)/(X_1 \times VP_1 + X_2 \times VP_2)$ **repeat Equation 6-1**

X_{DCS}	X_{TCS}	T,C	VP_{DCS}	VP_{TCS}	Y_{DCS}	Y_{TCS}
1.00	0.00	92.40	11.00	5.54	1.00	0.00
0.99	0.01	92.64	11.06	5.57	0.99	0.01
0.95	0.05	93.61	11.28	5.70	0.97	0.03
0.90	0.10	94.86	11.57	5.86	0.95	0.05
0.85	0.15	96.14	11.88	6.04	0.92	0.08
0.80	0.20	97.46	12.19	6.22	0.89	0.11
0.75	0.25	98.81	12.53	6.41	0.85	0.15
0.70	0.30	100.21	12.88	6.62	0.82	0.18
0.65	0.35	101.63	13.24	6.83	0.78	0.22
0.60	0.40	103.11	13.63	7.05	0.74	0.26
0.55	0.45	104.63	14.04	7.29	0.70	0.30
0.50	0.50	106.19	14.46	7.54	0.66	0.34
0.45	0.55	107.80	14.91	7.80	0.61	0.39
0.40	0.60	109.45	15.38	8.08	0.56	0.44
0.35	0.65	111.17	15.88	8.38	0.51	0.49
0.30	0.70	112.93	16.40	8.69	0.45	0.55
0.25	0.75	114.74	16.95	9.02	0.39	0.61
0.20	0.80	116.62	17.53	9.37	0.32	0.68
0.15	0.85	118.56	18.15	9.74	0.25	0.75
0.10	0.90	120.56	18.80	10.13	0.17	0.83
0.05	0.95	122.62	19.49	10.55	0.09	0.91
0.01	0.99	124.33	20.08	10.91	0.02	0.98
0.00	1.00	124.76	20.22	11.00	0.00	1.00

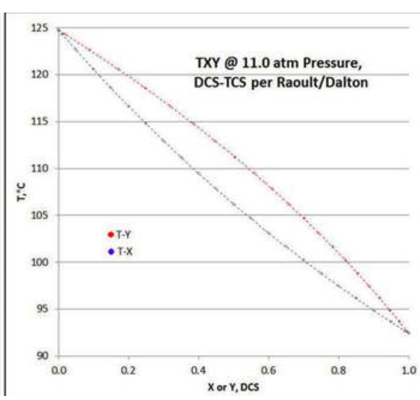


Figure 9-1, TXY calculation table and plot, Raoult/Dalton

If the data plot of Figure 9-1 plot is carefully examined, one can see that the T-**X** and T-**Y** curves are exactly symmetrical when “folded” along a line running from abscissa/ordinate points of (0,124.76) to (1,92.40); or “folded” orthogonally to that same line. In other words, there are no departures from ideality in either vapor or liquid phase, either with DCS or TCS.

Now **X** and **Y** are re-calculated including the departures from ideality, both in fugacity (ø) as well as Liquid Activity Coefficients (γ).

X_{DCS}	X_{TCS}	T,C	VP_{DCS}	VP_{TCS}	$ø_{L(DCS)}$	$ø_{V(DCS)}$	$ø_{L(TCS)}$	$ø_{V(TCS)}$	Υ_{DCS}	Υ_{TCS}	Y_{DCS}	Y_{TCS}
1.00	0.00	92.40	11.00	5.54	1.000	1.000	1.019	0.852	1.000	1.215	1.000	0.000
0.99	0.01	92.56	11.04	5.56	1.000	1.001	1.019	0.853	1.000	1.210	0.993	0.007
0.95	0.05	93.25	11.20	5.65	0.999	1.003	1.019	0.856	1.001	1.188	0.964	0.036
0.90	0.10	94.16	11.41	5.77	0.999	1.007	1.019	0.860	1.002	1.164	0.928	0.072
0.85	0.15	95.12	11.63	5.90	0.998	1.011	1.018	0.865	1.005	1.142	0.892	0.108
0.80	0.20	96.14	11.88	6.04	0.997	1.015	1.018	0.869	1.008	1.122	0.856	0.144
0.75	0.25	97.21	12.13	6.19	0.996	1.019	1.017	0.874	1.013	1.105	0.819	0.181
0.70	0.30	98.35	12.41	6.35	0.995	1.024	1.017	0.879	1.018	1.089	0.782	0.218
0.65	0.35	99.55	12.71	6.52	0.994	1.029	1.016	0.885	1.025	1.075	0.744	0.256
0.60	0.40	100.83	13.04	6.71	0.993	1.034	1.015	0.890	1.032	1.062	0.705	0.295
0.55	0.45	102.20	13.39	6.91	0.992	1.040	1.015	0.896	1.039	1.051	0.664	0.336
0.50	0.50	103.65	13.77	7.14	0.991	1.047	1.014	0.903	1.048	1.041	0.621	0.379
0.45	0.55	105.20	14.19	7.38	0.989	1.054	1.013	0.910	1.057	1.032	0.576	0.424
0.40	0.60	106.84	14.64	7.65	0.988	1.061	1.012	0.917	1.067	1.025	0.528	0.471
0.35	0.65	108.61	15.14	7.94	0.986	1.070	1.011	0.925	1.077	1.019	0.478	0.522
0.30	0.70	110.49	15.68	8.26	0.984	1.080	1.010	0.934	1.088	1.013	0.424	0.576
0.25	0.75	112.51	16.27	8.61	0.982	1.090	1.008	0.943	1.099	1.009	0.366	0.634
0.20	0.80	114.66	16.92	9.00	0.980	1.102	1.007	0.953	1.111	1.006	0.304	0.696
0.15	0.85	116.97	17.64	9.43	0.978	1.115	1.006	0.963	1.123	1.003	0.237	0.763
0.10	0.90	119.42	18.43	9.91	0.975	1.130	1.004	0.975	1.136	1.001	0.164	0.836
0.05	0.95	122.02	19.29	10.43	0.972	1.147	1.002	0.987	1.149	1.000	0.085	0.915
0.01	0.99	124.20	20.03	10.88	0.969	1.162	1.000	0.997	1.159	1.000	0.018	0.982
0.00	1.00	124.76	20.22	11.00	0.968	1.166	1.000	1.000	1.162	1.000	0.000	1.000

Figure 9-2, TXY/P calculation table, all departure factors included

The table in above Figure 9-2 is likewise set up and calculations performed row-wise for each value of $\mathbf{X}_{DCS}$, but the liquid and vapor-phase fugacity coefficients ($ø_L$ and $ø_V$) and activity coefficients (γ_i) are included in the calculation of vapor mole fractions (γ_{DCS}). For each row, the temperature is iterated until the sum of partial pressures = 11.0 atmospheres. Note that since the activity coefficients are mild functions of temperature, each row has its own Van Laar constants determined and the value of γ calculated for that row's liquid mole fraction. Thus all departure-from-ideality factors are included.

$$Y_i = (\phi_i^L \times \gamma_i \times VP_i \times X_i)/(\phi_i^V \times \pi) \tag{9.1}$$

which is an algebraic re-arrangement of Equation 6-4 from previous **Part VI**.

To better show how the several ideality departure factors impact the T**XY**/P plot (as compared to Figure 9-1 above), the table in Figure 9-2 is plotted. The symmetry of the Figure 9-1 Raoult/Dalton plot is no longer there, neither along the "fold" line running from abscissa/ordinate points of (0,124.76) to (1, 92.40); or "folded" orthogonally to that same line. Also note that the T-**X** and T-**Y** curves are closer together in Figure 9-3, as opposed to Figure 9-1.

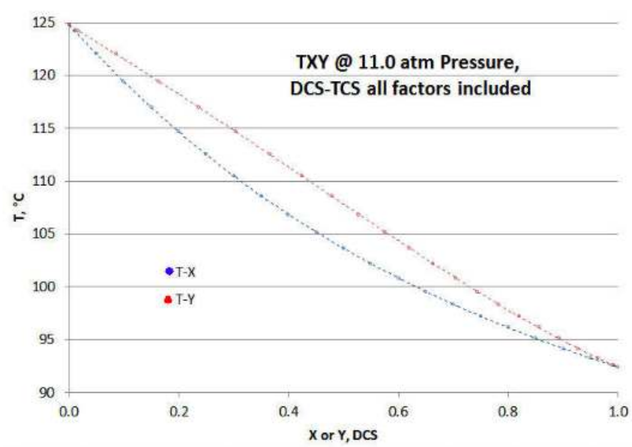


Figure 9-3, TXY/P plot, all departure factors included

For a very simple approximation of tray count, for given distillation column "tops" product , "bottoms" product and feed mixture mole fractions, and for a given reflux rate, a McCabe-Thiele plot is constructed in Figure 9-4 to graphically step off the number of trays above and below the feed tray. For almost all industrial applications, this tray-by-tray analysis is done via computer modeling, such as ASPEN, VMG, HYSYS, ChemCad, etc. To apply the technique to a computer model, the user would specify the critical constants (as per **Part III**), the vapor pressure equation to be used (as per **Part IV**), the EOS to be used to calculate fugacity coefficients (as per **Part V**), and the Liquid Activity Coefficient model to be used (as per **Part VI**).

To construct a McCabe-Thiele plot, the **X**-**Y** equilibrium (in red below) is plotted on a set of axes where **X**= liquid mole fraction of the most volatile component = $\mathbf{X}_{DCS}$; and **Y**= vapor mole fraction of the most volatile component = $\mathbf{Y}_{DCS}$. The total reflux line (in black below) connects (**X**,**Y**) = (0,0) and (1,1); representing the maximum reflux possible. The lines are added (in dashed blue below) for the feed ($\mathbf{X}_F$), and distillate product ($\mathbf{X}_D$) and the bottoms product ($\mathbf{X}_B$), from the **X**-axis to the total reflux line. The feed line is extended toward the equilibrium curve, and from that intersection a line is drawn to the end of the distillate product line. The slope of that drawn line represents the minimum internal reflux needed to make the separation (i.e., with an infinite number of trays). The upper column operating line (in magenta below) is then drawn connecting the end of the distillate product line with the feed line, with a slope somewhat greater than the minimum reflux slope. In normal practice the upper column operating line slope is usually about 1.3 times the minimum slope, depending on the relative value of operating energy vs capital cost. Then the lower column operating line is drawn, intersecting the bottoms product line with the upper column operating line (and the feed line). The number of theoretical trays (aka stages) can then be stepped off (in green below) from $\mathbf{X}_D$ to $\mathbf{X}_B$, and best tray determined for the feed.

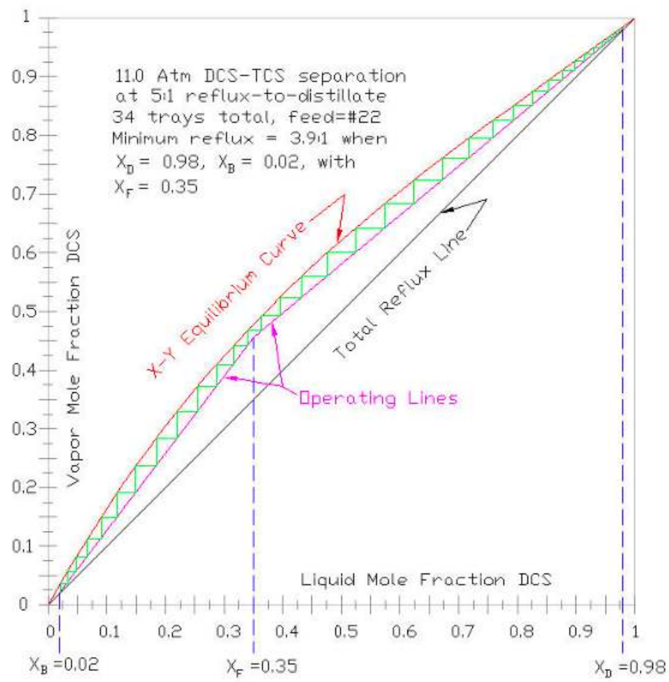


Figure 9-4, McCabe Thiele solution to example calculations

To illustrate an example of using a McCabe-Thiele plot, assume the mixture to be separated is a 35% DCS molar liquid, pre-heated to its boiling point of 108.61°C (as shown on the 35% of Figure 9-2). Further, assume that the column "tops" distillate product is to be 98% molar DCS and the column "bottoms" product is to be 98% molar TCS. Per the text above, the minimum reflux (i.e., the slope of the line between $X_D = X_{DCS}$ of 0.98, $X_B = X_{DCS}$ of 0.02) is determined to be 3.9:1, reflux-to-distillate. To allow a reasonable number of trays, a reflux rate of 5:1 is chosen (about 1.3 times the minimum), so the slope of the upper column operating line is $R/(R+1) = 5/(5+1) = 0.8333$. Stepping off trays, it is determined that the total number of theoretical trays required is 34, with the feed tray at #22. Conventional tray counting starts at the column condenser = tray #1.

As can be seen by examining Figure 9-4, as the column's reflux rate is increased, upper column (magenta) operating line's slope increases, and the gap opens up between the (red) X-Y Equilibrium Line and the (black) Total Reflux Line. So increasing the reflux decreases the number trays required to make the specified separation.

In a normal industrial process application, all of the calculations are done via computer simulators, allowing for variations in operating line slope; feed quality condition (e.g., saturated vapor, saturated liquid, sub-cooled liquid, etc); changes in the internal liquid/vapor flow ratios due to physical properties; ambient heat effects; tray efficiencies, etc; as well as integrating the distillation column into the rest of the process design. So the McCabe-Thiele method is just used for graphical explanation.

However, the primary consideration of the distillation column design is the X-Y equilibrium relationship. If this is considerably in error – such as using only Raoult/Dalton – the column design will be a failure. Given the cost of industrial-scale distillation systems being in the millions of dollars (tens of $MM for larger ones), there is every reason to get the X-Y equilibrium relationship correct.

In the example of this article, the basic Raoult/Dalton model would result in the X-Y equilibrium relationship in the above example having more curvature (than shown in Figure 9-4) and therefore more open gap between the equilibrium curve and operating lines. The resulting reflux determined would be about 2/3 of required, and number of trays required would also be about 2/3 of needs - so a major error in design. Given what is at stake financially, there is no question that the use of proper distillation science is well worth it, even if the concepts are complex and the calculations are tedious.

10: Convergence Strategy

Distillation Science (a blend of Chemistry and Chemical Engineering)

This is **Part X, Convergence Strategy** of a ten-part series of technical articles on Distillation Science, as is currently practiced on an industrial level. See also **Part I, Overview** for introductory comments, the scope of the article series, and nomenclature.

This last article discusses how to approach solving non-intrinsic equations and equation systems, since these are part of the practice of distillation science. While it can be argued that this topic is technically more related to math or computer science, the techniques are indispensable tools to obtaining workably accurate solutions.

Most relationships in Distillation Science feature equations that are written with temperature as the starting variable, and then one solves for the other parameters. For example, almost all vapor pressure relationships are written intrinsically with VP as a function of temperature. But frequently the problem statement begins with a fixed pressure, and so there needs to be an iterative procedure to solve for temperature.

More complex relationships involving two starting variables are written in temperature and pressure, with the equation's dependent variable written as a function of (absolute) temperature and (absolute) pressure. An example is an Equation of State, where Z is written as f(T,P), with molar volume (V) being back-solved from the defining relationship PV=ZRT. Some scientific relationships are written in dimensionless "reduced" variables, where T, P, and V are divided by their critical point value (e.g., $T_r=T/T_c$, $P_r=P/P_c$), which allows for more global equation forms. The Peng-Robinson Equation of State is an example of using reduced properties, although Equation 5-4 partially masks that by breaking the equation into pieces.

The most basic technique for solution convergence in a non-intrinsic relationship is based on "gaming strategy": (1) pick a possible independent variable's value and work it through to a solution, comparing it to the desired final dependent variable; then (2) move the value choice a little bit and see how that helps. This is just a methodical form of "trial and error", and will eventually lead to a solution. However, such strategy is almost impossible to automate for easy programming, as in a spreadsheet macro or a stand-alone BASIC program. Moreover, it can be hard to get started, may lead to an unstable situation that cycles around a solution rather than stably converging to the required accuracy.

Newton-Raphson Method

A better convergence option is a modification of Newton-Raphson, aka Newton's Method. In the instructions below, "X" is the relationship's independent variable and "Y" is the dependent variable. The problem is that you have a desired target value for "Y" and have to iteratively seek the correct value for "X", and there is no way to re-write the relationship so that "Y" is the independent variable. The increasing subscripts denote the progression at improving the solution, such that "Y_n" is identical to the desired target value of "Y" = "Y_T", and "X_n" is the value for "X" that yields that solution.

1. If there is a way to get a very rough estimate of solution, start with that as the first trial = X_0, and work the scientific relationship through to a first possible solution, Y_0. (If there is no rough estimation possible, start with a known "safe" initial X_0 and calculate Y_0.) Then make an extremely small ΔX step (say 0.1% change in X_0). In the next iteration, $X_1=X_0+\Delta X$ would be evaluated to get Y_1. If the change in ΔY (i.e., $\Delta Y= Y_1-Y_0$) is too inconsequential to be seen, increase the value of the ΔX step, until there is some noticeable change in ΔY. (This is an important consideration since some relationships are complex, so it is hard to pick that first ΔX step to get a reasonable change in Y). It does not matter whether the change to X (i.e., ΔX) is positive or negative: that will get worked out later.
2. Calculate the quantity $m=(\Delta Y/\Delta X)=(Y_1-Y_0)/(X_1-X_0)$, which approximates the scientific relationship's first derivative. "m" can be positive or negative (which is why the sign in ΔX didn't matter). A positive value for "m" indicates that Y increases with increased X; or a negative value for "m" indicates the opposite effect. Take notice of the sign of "m", to see if its sign makes common sense.
3. With Y_T as the target solution value, make the next guess for X as

$$X_2=X_1+K\times(Y_T-Y_1)/m \tag{10.1}$$

4. The variable "K" is the relaxation factor, with values of "K" closer to unity giving faster convergence, but at the peril of overshoot or convergence instability; and "K" values closer to zero giving slow but stable convergence. Until you know something about the scientific relationship, it is recommended using a "K" value of about 0.2-0.3 at first, especially when

logarithmic relationships are involved. You can always change the “K” value once you see how sensitive the scientific relationship is.

5. For the “*i*th” iteration, the iteration equation becomes:

$$X_{i+1} = X_i + K \times (Y_T - Y_i) \times (X_i - X_{i-1})/(Y_i - Y_{i-1}) \tag{10.2}$$

6. Continue iterating "n" times until the solution, Y_n, is adequately close to the target solution, Y_T. It is suggested that loop iteration accuracy should be one decimal place greater than that of the final solution’s desired precision, to allow for round-off error and computer calculation precision.

When doing automated computer calculations, either via spreadsheet macro or stand-alone BASIC program, it is a good idea to error-check each iteration for division by zero, and to limit the number loop iterations (say to a high number like 100). Otherwise, the iteration program can get “hung up” in an infinite loop. If you are getting essentially zero value for the quantity $(Y_n - Y_{n-1})$ or requiring an excessive number of loops to converge, that requires adjusting the relaxation factor, “K”, up or down to adjust convergence sensitivity.

Some ideas for making that first rough estimate, to get the iteration started, are:

- For finding the saturation temperature for a known pressure (i.e., inverting a vapor pressure relationship), start with using a basic VP relationship (inaccurate, but simple), such as $Ln(P)\alpha - \Delta H_{vap}/RT$ using the known ambient boiling temperature, T_b @ P=1 atmosphere and a reasonable estimate for latent of vaporization,ΔH_{vap}. Then the rough starting estimate is $T_0 = T_b - \Delta H/[R \times Ln(P_T)]$, where P_T is the known pressure target, in atmospheres. This will give a high estimated T_0 for $P_T >$ atmospheric, and a low estimated T_0 for $P_T <$ atmospheric. It is more important to start with a "safe" estimate than close first iteration, to avoid the pitfalls of automated solutions mentioned above.
- Vapor pressure charts and tables are good ways to get started, although sometimes hard to program into an automated sequence.
- When iterating for the saturation temperature of a mixture, start with the assumption of a linear relationship between the two fluids’ saturation temperatures at that pressure; then interpolate between those two pure-component temperatures per the mixture’s mole fraction. There is always curvature between two fluids’ saturation temperatures, so assuming linearity isn’t correct; but it gets the iteration loop started.
- Starting an iteration loop for fugacity coefficients should only be done after a cursory look at the reduced pressure, P_r, since these relationships involve calculating compressibility, Z. For liquid fugacity coefficients, avoid starting out or getting too close to P_r=0.9, because the automated solution to cubic equations gets too sensitive, and one of the roots may become an “imaginary number” (i.e., a number times the square root of one). For vapor fugacity coefficients, do an initial check to see which Z = f(T_r, P_r) equation is best to use (i.e., *Equation 6-6* or *Equation 6-7*): one is better for lower values of P_r and one better for higher values. If you attempt to cross over between equations in the middle of a solution, that discontinuity might cause an automated iteration to “blow up”.
- Resist the temptation to use Newton-Raphson to get the roots for the cubic relationship Z: there are three roots, and only two are scientifically meaningful. Besides, you need to know all three roots in order to judge which is Z_v (the largest of the three) and which is Z_L (the smallest of the three). Instead, go through the procedure in **Part V** and solve the cubic equation for Z, using the exact trigonometric method of *Equation 5-4*.
- Starting the loop for activity coefficients should never be done at either extreme of mole fractions (zero or unity). The Liquid Activity Coefficient's relationships are complex-shaped and can have $\gamma/\Delta X$ values that are almost zero (near **X**=1) or very high (near **X**=0). For bulk separations, start somewhere between 0.3<**X**<0.7, and use a moderate value for relaxation factor, “K” (say 1/3) in Equation (10.2) . For trace impurity distillation systems, either start closer to **X**=0.1 or at **X**=0.9, plus use a conservative relaxation parameter, say K=0.1.

Convergence Strategy for modified Thek-Stiel VP Equations

The “sting” of working out nested loop iterations has been removed by giving the solutions for the modified Thek-Stiel VP equation (Equation 4-3) in tabular form (Table 4-1). Obtaining vapor pressure solutions for fluids other than chlorosilanes and their impurities (see **Part IV)** require more complex convergence strategies than Newton-Raphson.

These VP solutions involve a system of nested-loops: the value of the Watson coefficient “q” depends on evaluating the candidate vapor pressure equation at T_r=0.7 to get the loop’s estimate of acentric factor; and obtaining the loop’s estimate of “c” and “n” depends on finding a value of α_c that gives the correct saturation temperature at ambient pressure (aka, the normal boiling point).

The following solution strategy is advised below, *for fluids other than those of Table 4-1 (or in case one wants to check the table's values)*.

First, be sure all of the values for T_b, T_c, P_c are well-validated, and the degree of hydrogen bonding is known, per the definitions in **Part IV**. If the new fluid to be investigated is of a periodical table group other than III, IV, or V, the solution will involve a more intricate solution than below, since "k" (i.e., the degree of hydrogen bonding) is not well determined for other than Groups III-V, and a double iteration will be needed. Also be sure the vapor pressure data is also well-validated. A good check on that is to use a simplistic extrapolation of the known VP data, to see if the rough value of the acentric factor matches with the values of T_c and P_c. See **Part III** for more details on acentric factor, and especially Equation 3-3 For example, if an extrapolation of VP data using the Antoine VP relationship (Equation 2-3) in conjunction with values for T_c and P_c , ends up giving an acentric factor value over 0.3 for a compact molecule, or an acentric factor under 0.1 for a complex molecule, something is wrong with the data (either the VP data or the values of T_c and P_c).

For Equation 4-3, the value of the hydrogen bonding parameter, k, is established by experimental data for Group III, IV, and V hydrides, chlorides, oxychlorides, intermediate hydrochlorides, and several alphatic hydrocarbons. The value of k ranges from 0.12 to zero, and is tabulated in Part IV for many fluids. The rules for determining "k" are a bit complex, and are available upon request from the author. In brief, for Group III, IV, and V hydrides, k=0.1197; 0.0914; and 0.1009 respectively. For all fully chlorinated or methylated compounds of Groups III, IV, and V , k = 0.0291. For aliphatic hydrocarbons, k = 0. For other hydrochloride intermediates, a quadratic function has been determined to interpolate between all-hydride and all-chloride. These rules for "k" values are used in the entries of Table 4-3.

Having a rough estimate of the ambient pressure latent heat of vaporization is handy (but not necessary) for the first guess at the T-S heat variable. If no estimate is available, consult one of the many handbooks of chemical compounds, such as Yaw's *Chemical Property Handbook*, or Lange's *Handbook of Chemistry* and look for a latent heat value for a compound of similar complexity. It is always better to start a new fluid's solution to the Thek-Stiel VP equation with a low estimate for the T-S heat variable, and then increase.

The overall solution convergence strategy to the modified Thek-Stiel VP equation involves a triple-nested iteration loop, with the logic flow sheet of the strategy shown on Figure 10-1. The solution is obtained by improving candidate solutions for the T-S heat variable, the acentric factor (ω), and the α_c parameter, until all loops are converged.

The inner-most iteration loop is the α_c parameter (a thermodynamic consistency parameter from **Part IV**) , which is adjusted until the VP equation gives the correct saturation temperature at atmospheric pressure, aka the normal boiling point T_b. That loop is started by using Reidel's estimation method: calculating the rough value of $\psi_b = (-35 + 36/T_{br} + 42LnT_{br} - T_{br}^6$, and then plugging in that rough value of ψ_b in the equation below:

$$\alpha_c \approx (0.315\psi_b + LnP_c)/(0.838\psi_b + LnT_{rb}) \tag{10.3}$$

The middle of the nested loops is the acentric factor, which can be conveniently started using the Lee-Kister acentric factor (ω) approximation (which is not very accurate, but a good loop-starting estimate):

The Lee-Kister approximation for acentric factor (ω) conveniently uses only P_c and $T_{br}=T_b/T_c$:

$$\omega \cong \frac{-lnP_c + A + B/T_{br} + C * LnT_{br} + D * T_{br}^6}{E + F/T_{br} + GLnT_{br} + H * T_{br}^6} \tag{10.4}$$

A = -5.92714 E = 15.2518

B = 6.09648 F = -15.6875

C = 1.28862 G = -13.4721

D = -0.169347 H=0.43577

Based on this rough estimated value of ω, a value of *Watson's coefficient* "q" (from Equation 4-3) can be determined per the relationship

$$q = 0.37028 + 0.065404 * \omega \tag{10.5}$$

The outer-most loop is the T-S heat variable, "T-S ΔH_v", which doesn't really have much scientific meaning, but is a part of the Thek-Stiel relationship. This parameter is iterated based on the error function comparing the VP equation's output against the various known (or estimated) VP data points. To get the best overall fit, I suggest including data near T_r = 0.7 (where the acentric factor is evaluated). In getting the various T-S VP Equation solutions shown on Table 4-1, I chose to not automate this outer loop, but to rely on the more basic "gaming strategy" ⇒ always starting low, so as to monitor the convergence better. For computer automation, I used a BASIC program (actual software used was Liberty Basic version 4.03, selected due to prior familiarity). The starting guess for the outer loop iteration was based on the atmospheric ΔH_{vap} given in chemistry handbooks and from internet sources, or based on similar compounds where no published estimate of atmospheric ΔH_{vap} was available. This choice of starting guess was based on the "safe guess" criterion, and opposed to a scientific estimation.

Figure 10-1 below schematically gives the convergence strategy previously used with success. After insuring that all input parameter and data are reasonable, start with the initial guess for the T-S heat variable, "T-S ΔH_v". Then estimate a starting ω per Equation (10.4); a from that estimate a starting α_c per Equation 10-3. Now calculate values for constants "A", "B_0-B_3", "c", and "n" in Equation 4-3, and see if the inner-most loop's convergence is satisfied. If not, iterate the value of α_c until the correct VP is shown at T_b of 1.0 atmosphere (i.e., the normal boiling point). This inner-most loop is best converged by approaching the VP @ T_b from P<1 atmosphere, with convergence to ± 1.0E-4 atmospheres being seen within about ten iterations via Newton-Raphson, using a relaxation factor of 0.4 in Equation (10.2). Note that if the first guess of α_c shows a VP @ T_b of >1.0 atmospheres, just keep reducing the value of α_c by 10% until the convergence approach is from lower values of T_b. Trying to converge from the "high side" can be unstable.

Once the inner-most loop is converged, start changing the value of the acentric factor (ω) to converge the middle loop. This loop converges quite easily with just simple continuous substitution. Newton-Raphson is not required. Just put in the last value of ω, calculate the candidate VP constants ("A", "B_0-B_3", "c", and "n" in Equation 4-3) and determine a new value for ω based on the VP at T_r=0.7 (see Equation 3-3 for the definition of ω).

The outer loop is converged by picking that value of the T-S heat variable, "T-S ΔH_v", that best satisfies the error function of the VP data (i.e., predicted VP's as compared to the known data point VP's). I found the best overall fit used a least squares approach which emphasized the data near T_r=0.7, and which compared the VP equation error on a relative level (as opposed to an absolute level). If absolute VP equation errors are used (as opposed to relative ones), that over-emphasizes data at higher temperatures.

Index

Glossary

Sample Word 1 | Sample Definition 1

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